

ECEN721: Optical Interconnects Circuits and Systems Spring 2026

Lecture 1: Introduction



Sam Palermo
Analog & Mixed-Signal Center
Texas A&M University

Announcements

- Homework 1 is posted on website and due Jan 20

Class Topics

- System and design issues relevant to high-speed optical interconnects
- Channel properties
 - Modeling, measurements, communication techniques
- Optical interconnect circuits
 - Drivers, receivers, equalizers, timing systems
- Optical interconnect system design
 - Modeling and performance metrics
- Optical interconnect system examples

Administrative

- Instructor:
 - Sam Palermo
 - 318-E WEB, 979-458-4114, spalermo@tamu.edu
 - Office hours: MW 8:00AM-9:30AM
 - In-person and via Zoom
- Lectures: TR 3:55PM-5:10PM, ETB 1003
- Class web page
 - <https://people.engr.tamu.edu/spalermo/ecen721.html>

Class Material

- Textbook: Class Notes and Technical Papers
- Key References
 - *Broadband Circuits for Optical Fiber Communication*, E. Sackinger, Wiley, 2005.
 - <http://onlinelibrary.wiley.com/book/10.1002/0471726400>
 - *Design of Integrated Circuits for Optical Communications*, B. Razavi, McGraw-Hill, 2003.
 - *Advanced Signal Integrity for High-Speed Digital Designs*, S. H. Hall and H. L. Heck, John Wiley & Sons, 2009.
 - *High-Speed Digital Design: A Handbook of Black Magic*, H. Johnson & M. Graham, Prentice Hall, 1993.
- Class notes will be posted on the web

Grading

- Exams (50%)
 - Two midterm exams (25% each)
- Homework (25%)
 - Collaboration is allowed, but independent simulations and write-ups
 - Need to setup CADENCE simulation environment
 - Turn in via Canvas
 - No late homework will be graded
- Final Project (25%)
 - Groups of 1-3 students
 - Report required
 - Turn in report via Canvas

Prerequisites

- This is a circuits & systems & photonics class
- Circuits
 - ECEN474/704 or approval of instructor
 - Basic knowledge of CMOS gates, flops, etc...
 - Circuit simulation experience (HSPICE, Spectre)
- Systems
 - Basic knowledge of s- and z-transforms
 - Basic digital communication knowledge
 - MATLAB experience
- If you are strong in photonics, but weak in the above areas, then the assignments can be adjusted for more of a photonics emphasis

Simulation Tools

- Matlab
- ADS (Statistical BER link analysis)
- Cadence
- 90nm CMOS device models
 - Can use other technology models if they are a 90nm or more advanced CMOS node
- Other tools, schematic, layout, etc... are optional

Schedule

Topic		Week
I.	Optical Channel Properties	Week 1-8
II.	Optical Devices	
III.	Receiver Analysis	
IV.	Receiver Circuits	
1st MIDTERM		Mar. 3
V.	Transmitter Analysis	Week 9-14
VI.	Transmitter Circuits	
VII.	Laser Sources	
VIII.	RF Photonics & Photonic NoCs	
Project Report Due		Apr. 28
2nd Exam		May 4 (1:00PM – 3:00PM)

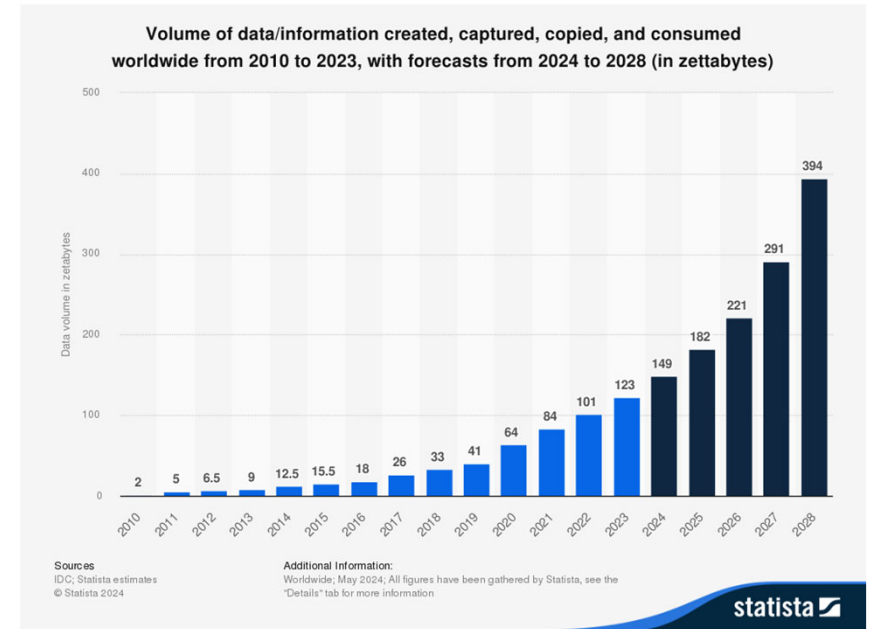
Optical Interconnects

- Electrical Channel Issues
- Optical Channel
- Optical Transmitter Technology
- Optical Receiver Technology
- Optical Integration Approaches

Exponentially Increasing Data Demands

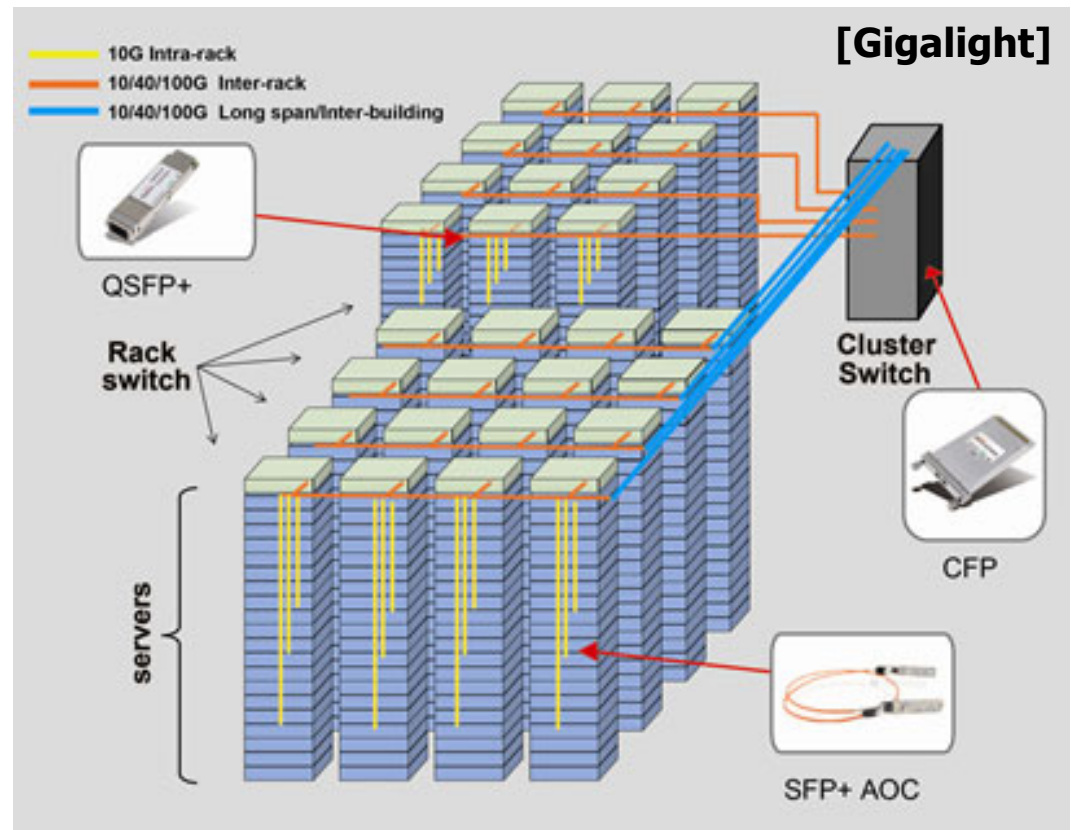
- Paradigm shift in data generation and processing
- Increased data generation
 - Artificial intelligence
 - Remote work/entertainment
 - Mobile communication
- Growing machine-to-machine traffic
- Hyperscale data centers are being optimized to infer intelligence from large data sets

Worldwide Data Volume (Zettabytes)

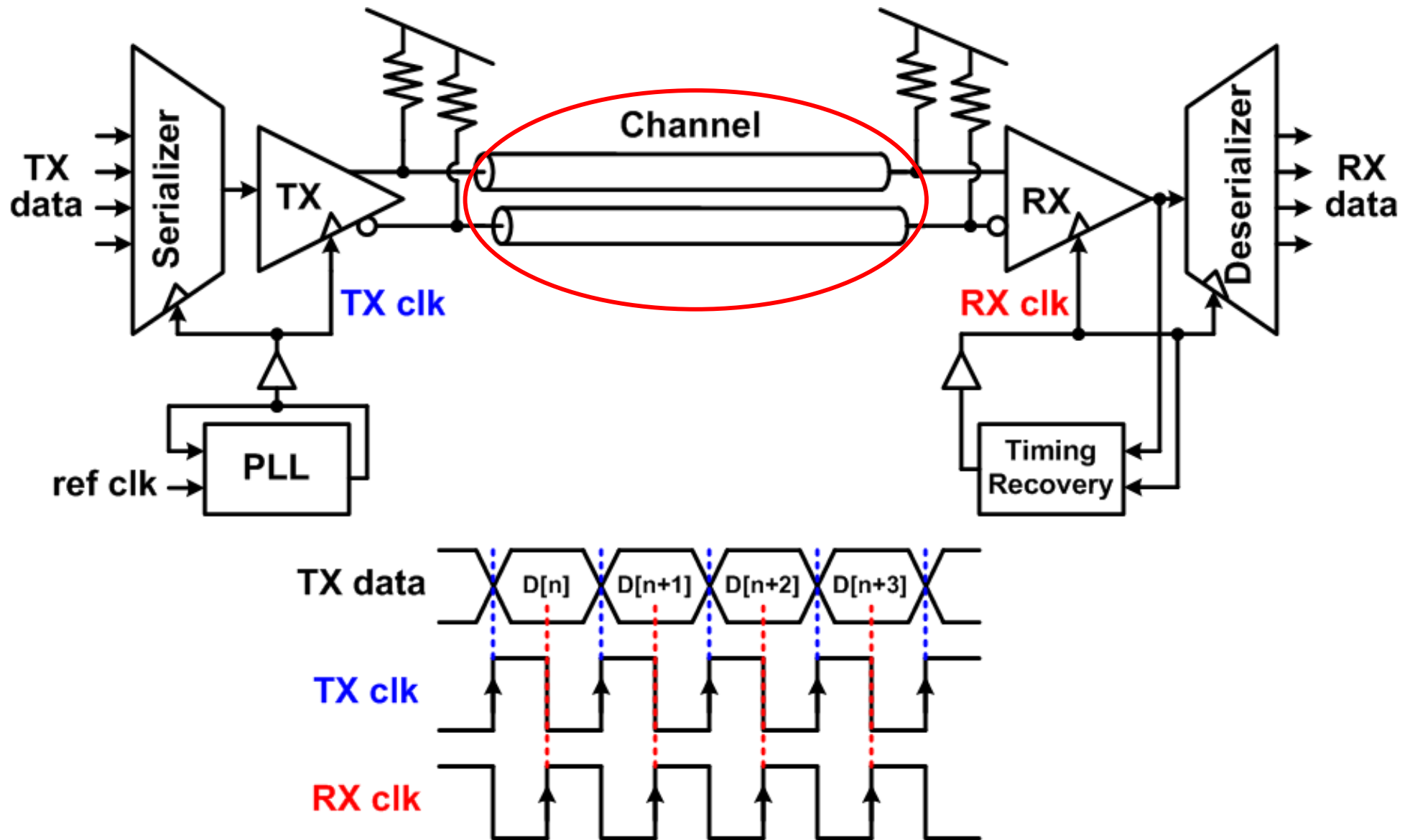


Data Center Links

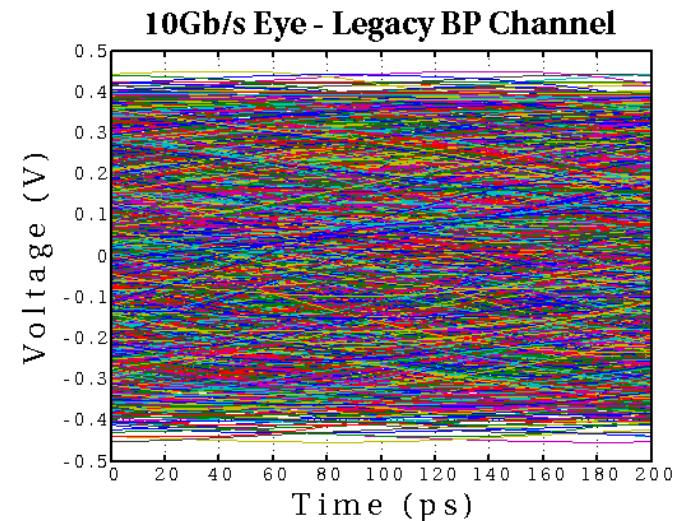
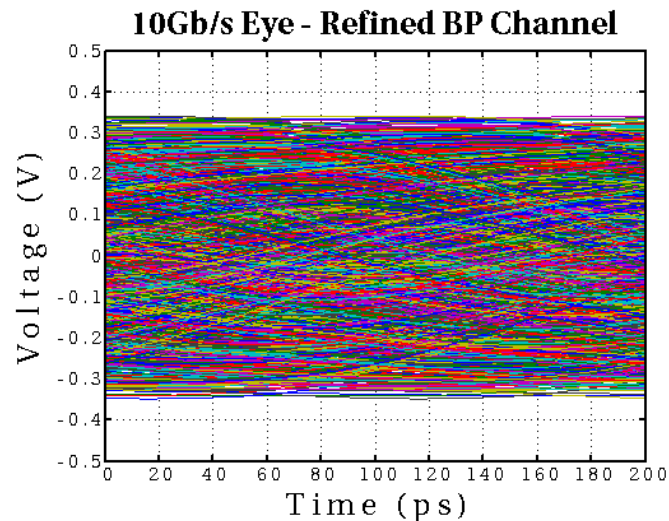
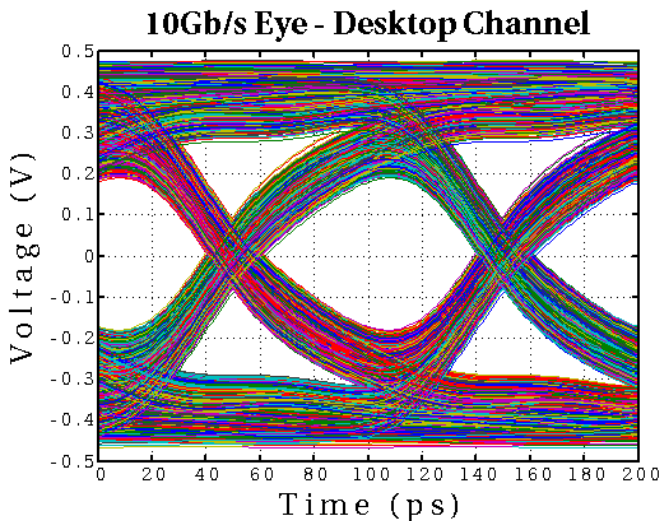
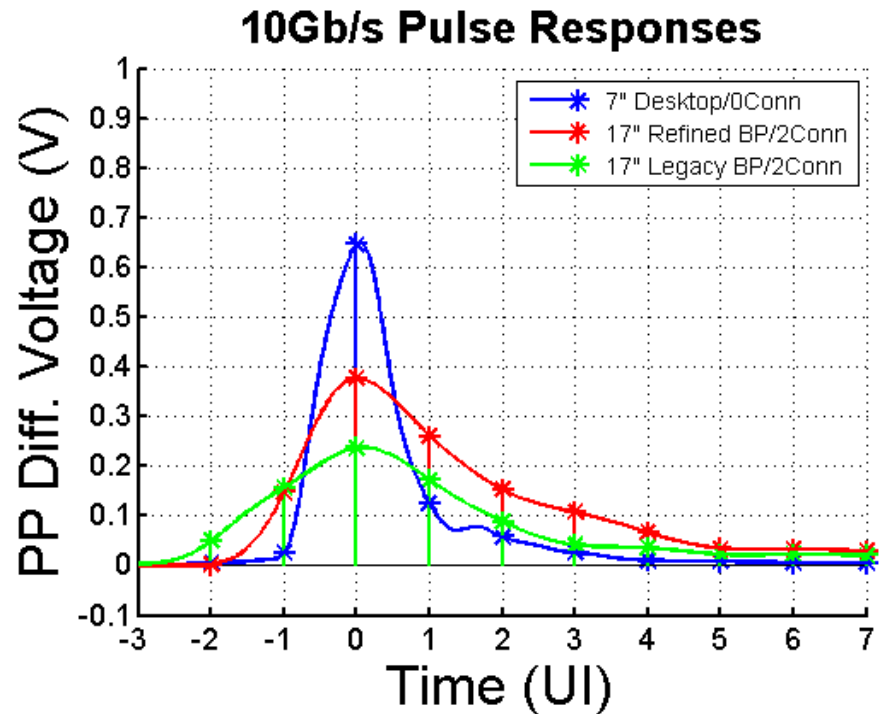
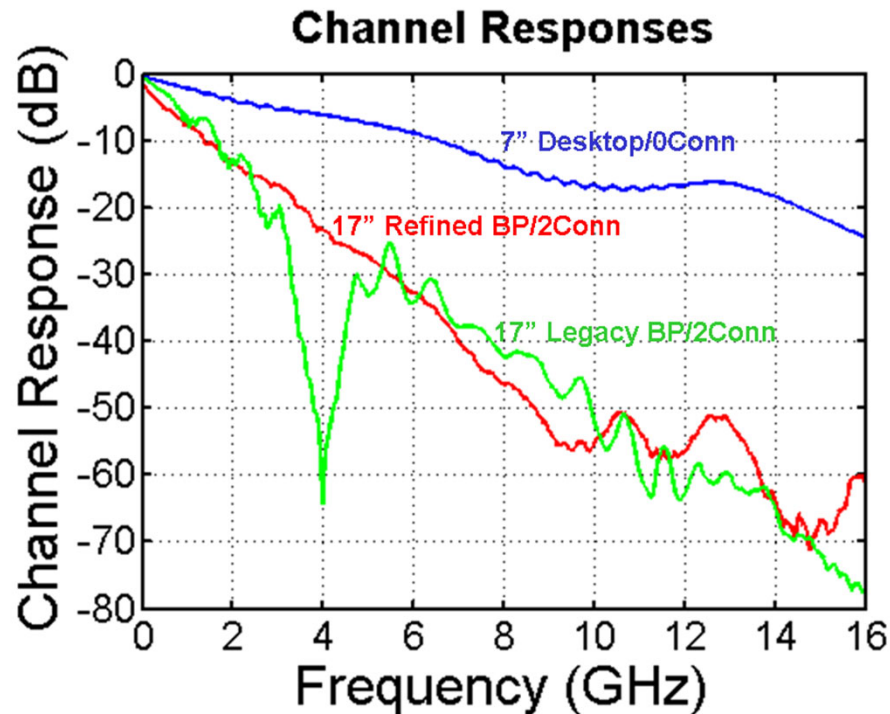
- Different interconnect technologies are used to span various distances
- Electrical I/O
 - Chip-to-module
 - Intra-rack (DAC cables)
- Optical I/O
 - Intra-rack (AO cables)
 - TOR switch to edge switch



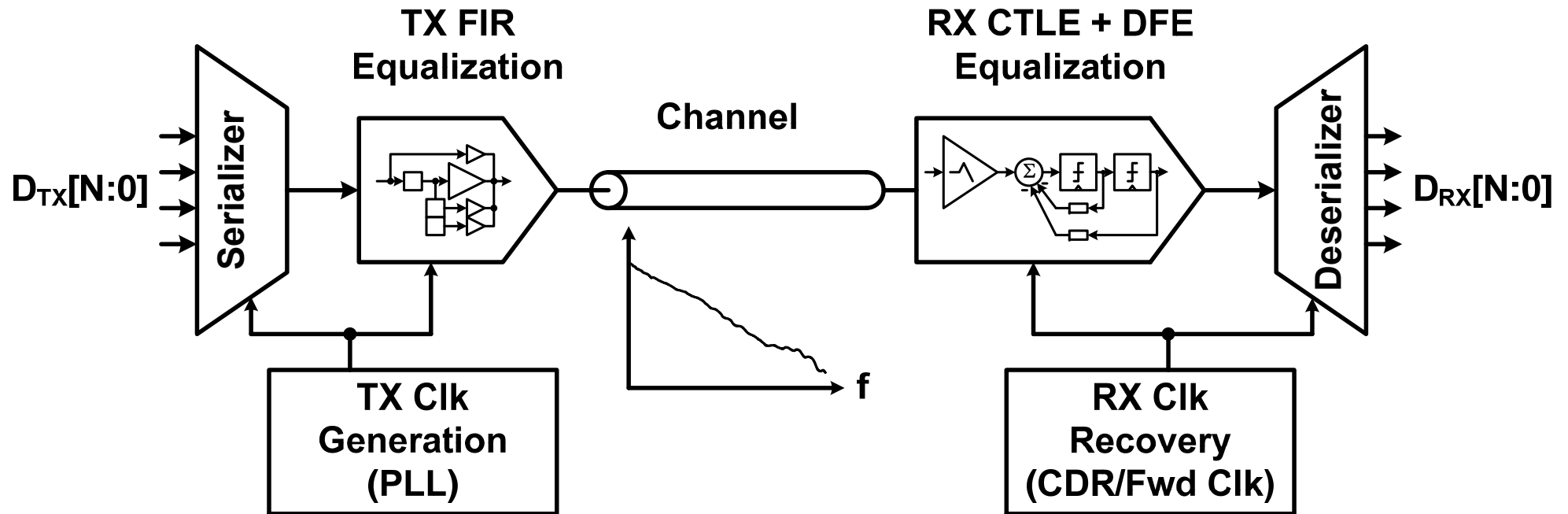
High-Speed Electrical Link System



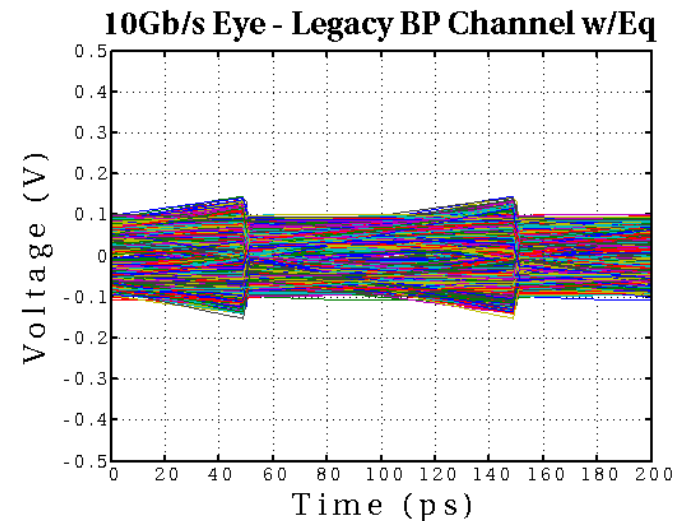
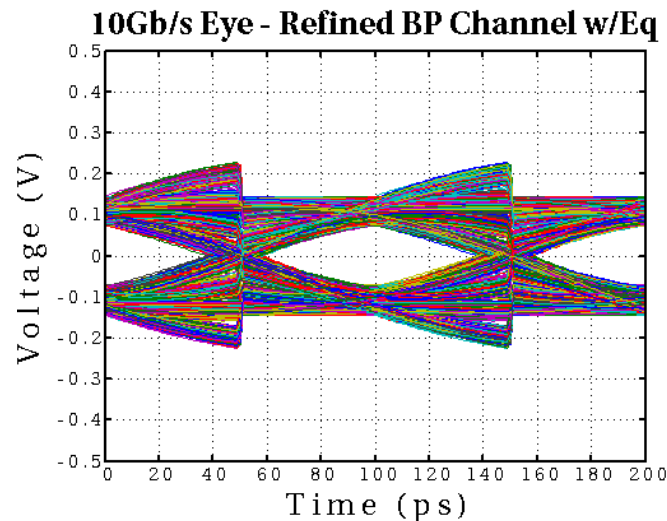
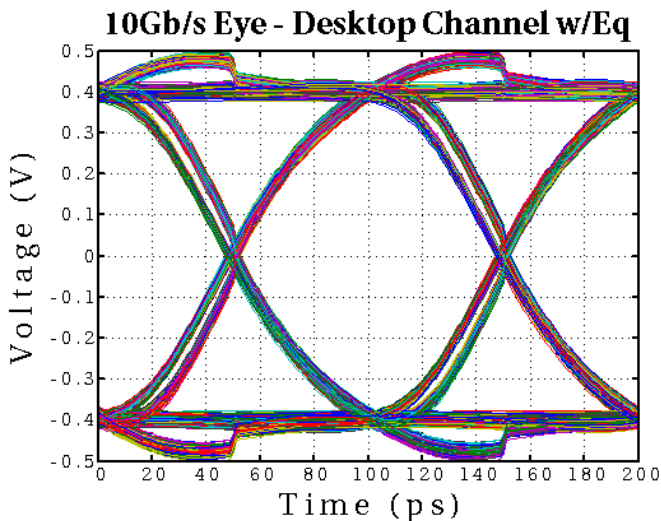
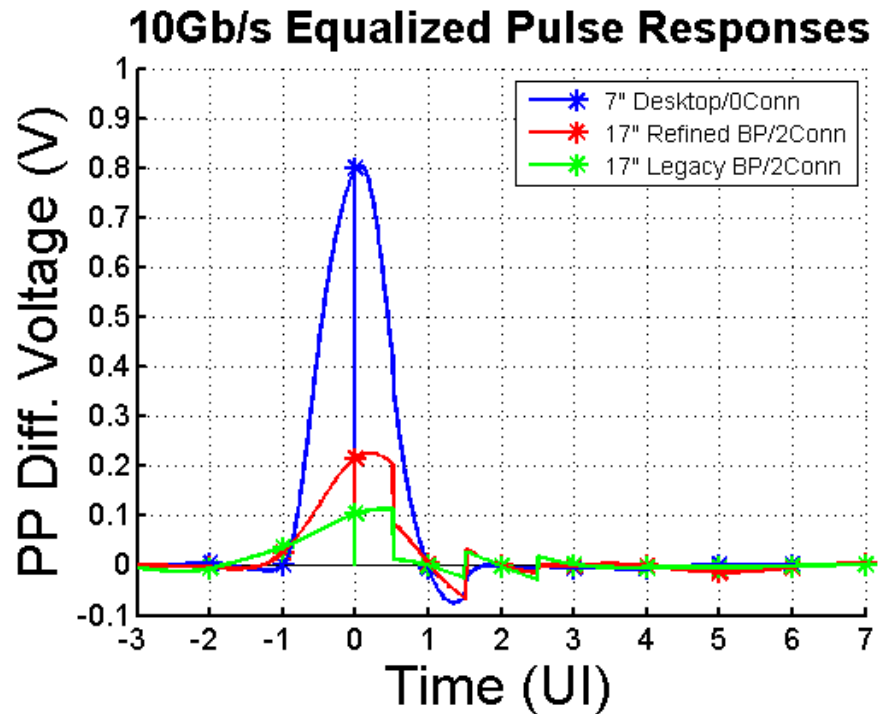
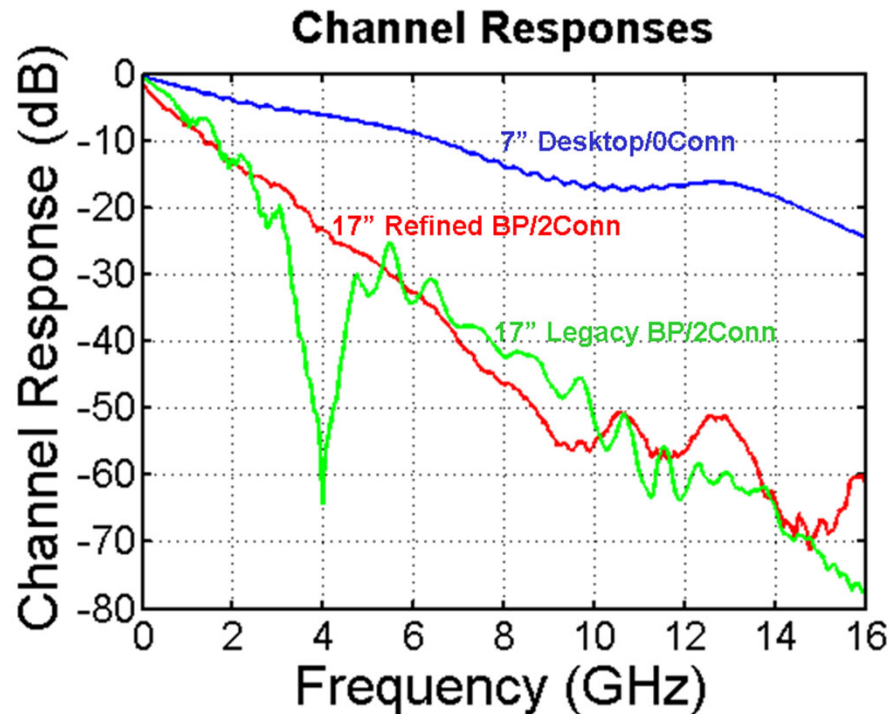
Channel Performance Impact



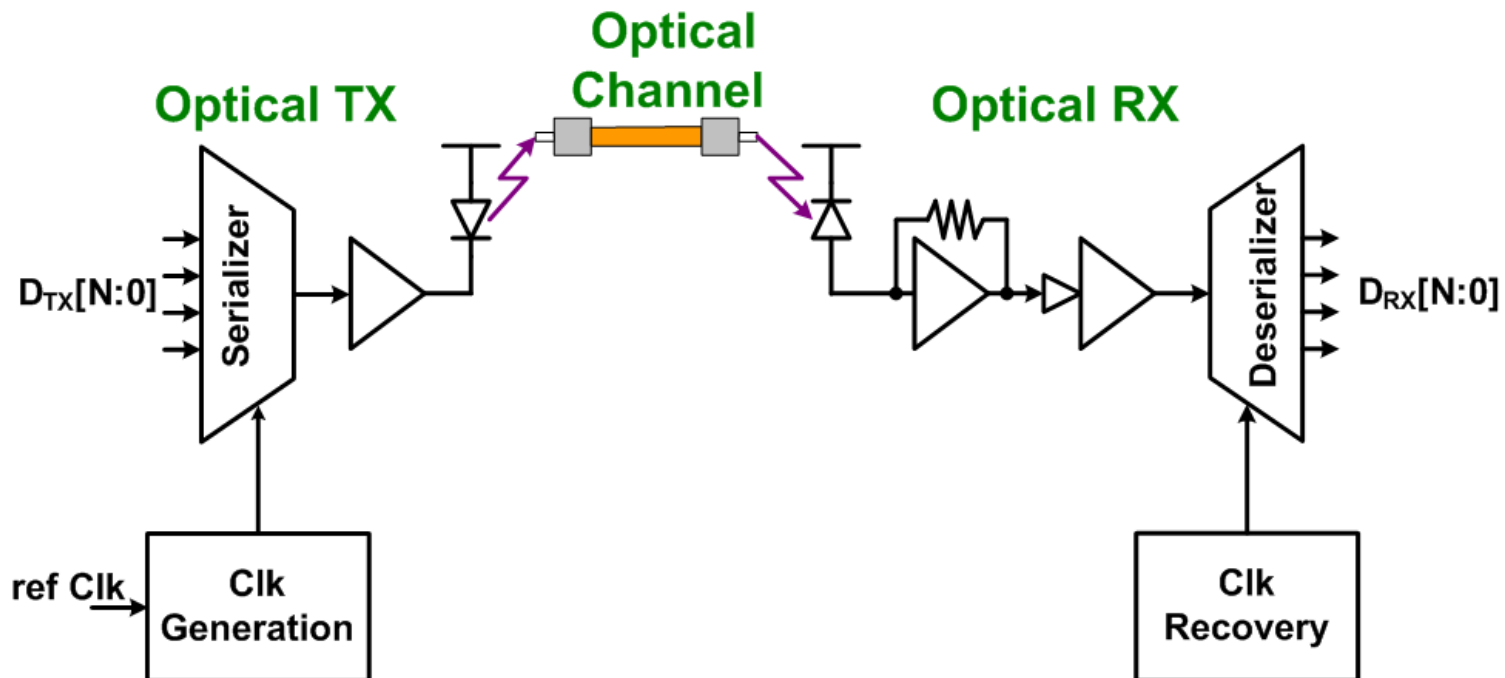
Link with Equalization



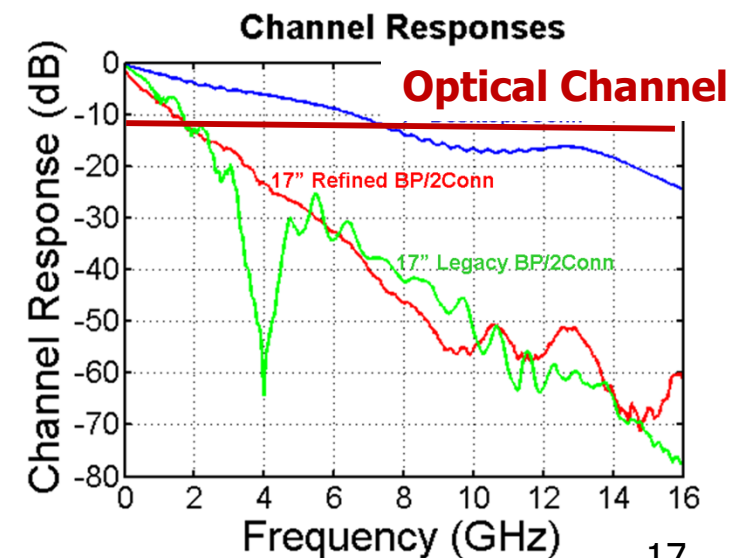
Channel Performance Impact



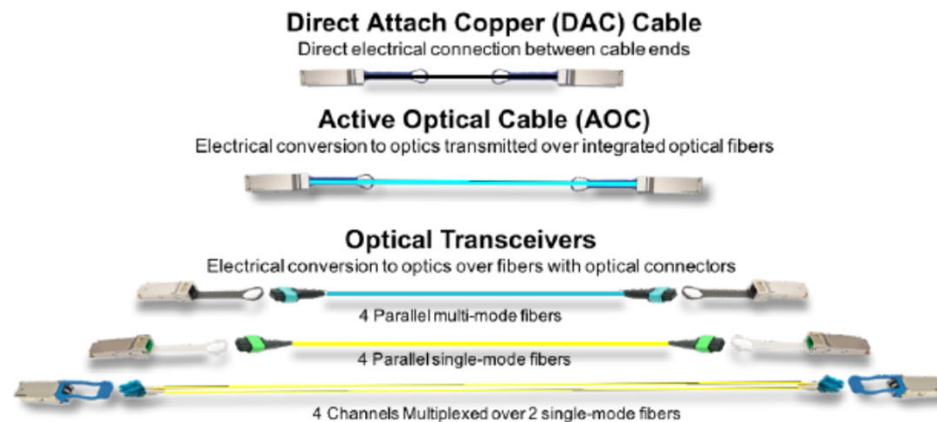
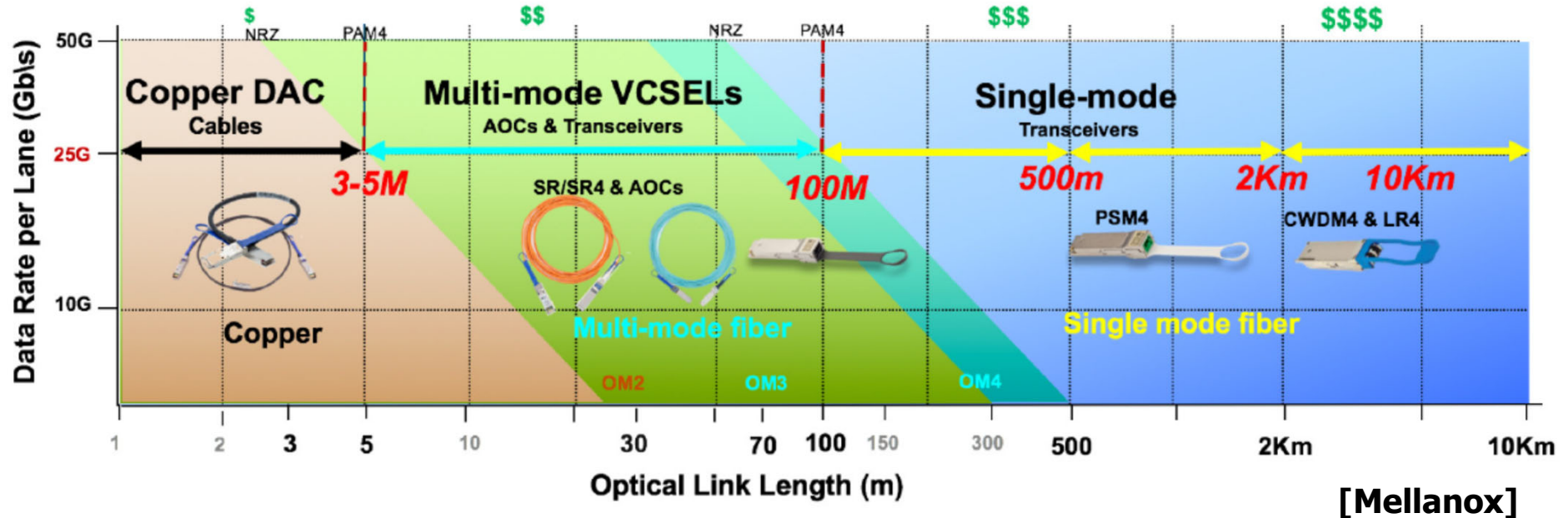
High-Speed Optical Link System



- Optical interconnects remove many channel limitations
 - Reduced complexity and power consumption
 - Potential for high information density with wavelength-division multiplexing (WDM)



Data Center Link Length

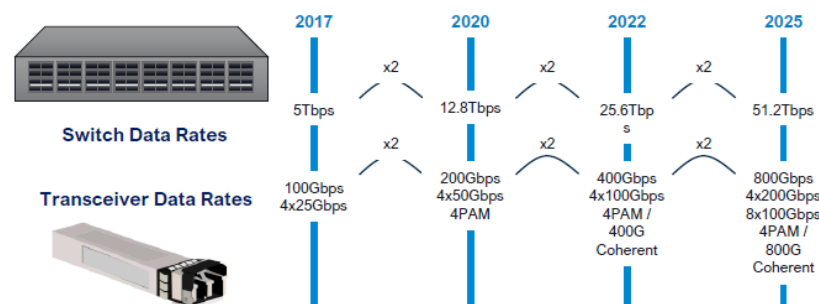
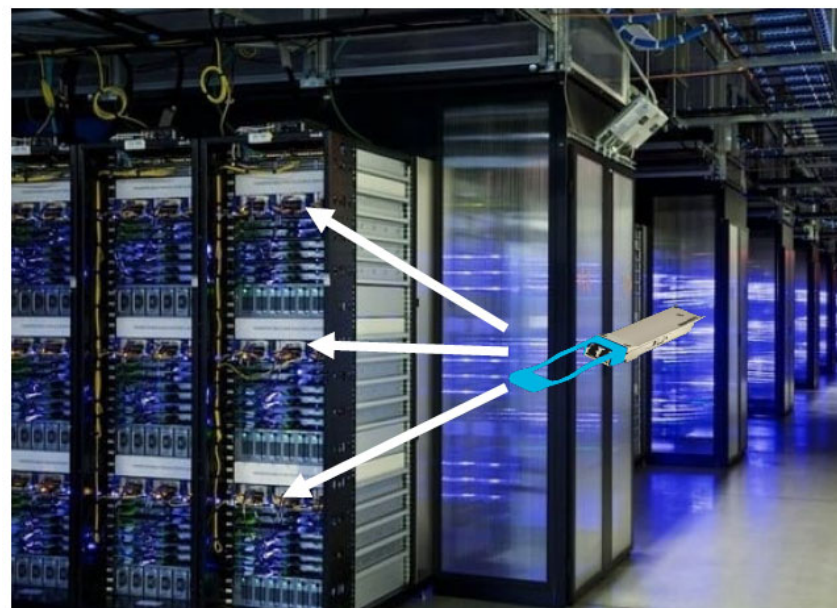


- Maximum reach scales inversely with data rate

AI Datacenter Connectivity Demands

- Higher per-lane data rates
- Scaling up with more xPUs in compute clusters
- Scaling out to larger clusters
- Flatter network topologies with higher radix switches to reduce latency
- Pushing optical connectivity to within the rack
- Optical modules often consist of 4 or 8 transceiver channels

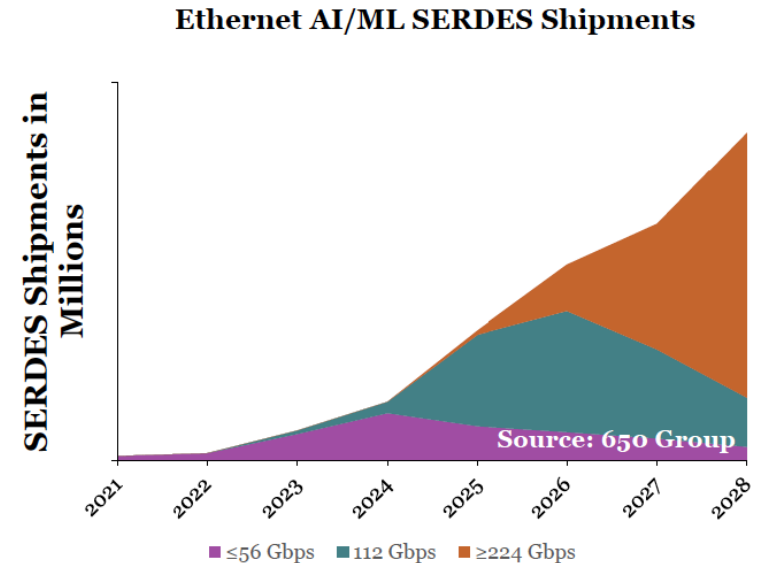
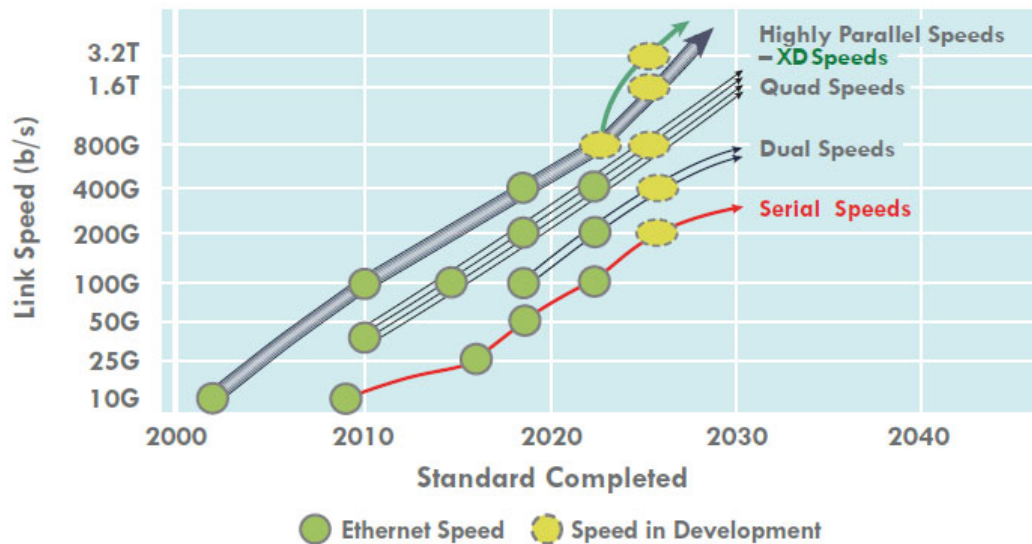
[Wang ISSCC 2024]



[Chan Carusone ISSCC 2024]

Ethernet Roadmap

<https://ethernetalliance.org/>



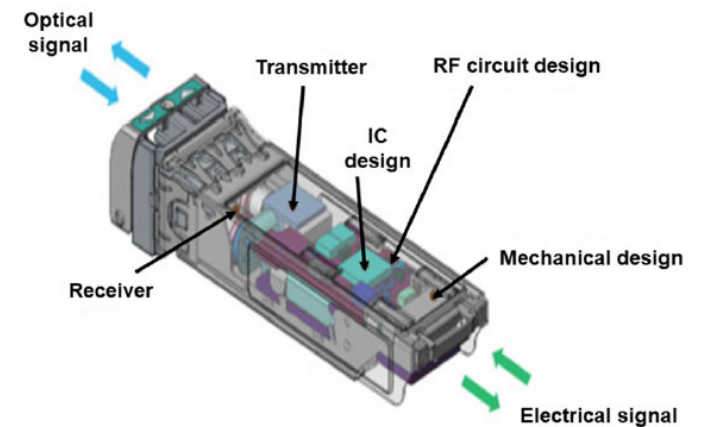
- Ethernet transceivers up to 3.2Tb/s in development
- Translates to high demand for 224Gb/s SERDES

Optical Modules

- Retimed pluggable modules
 - TX/RX DSP in advanced node CMOS
 - Separate driver and TIA chips, which can be SiGe, InP, CMOS
- Linear optics
 - Remove DSP to save power
 - Need to be very linear and have high bandwidth

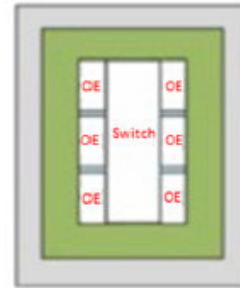


[Lau JEP 2025]

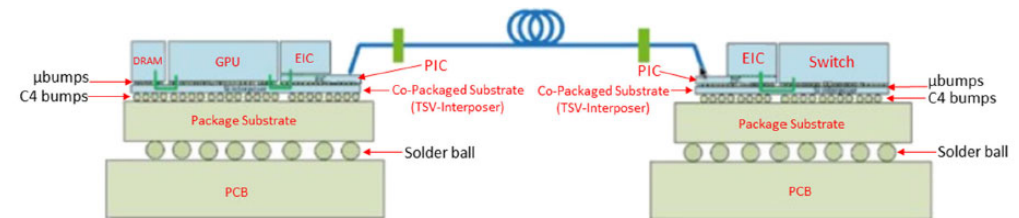


NVIDIA CPO Example

[NVIDIA GTC 2025]

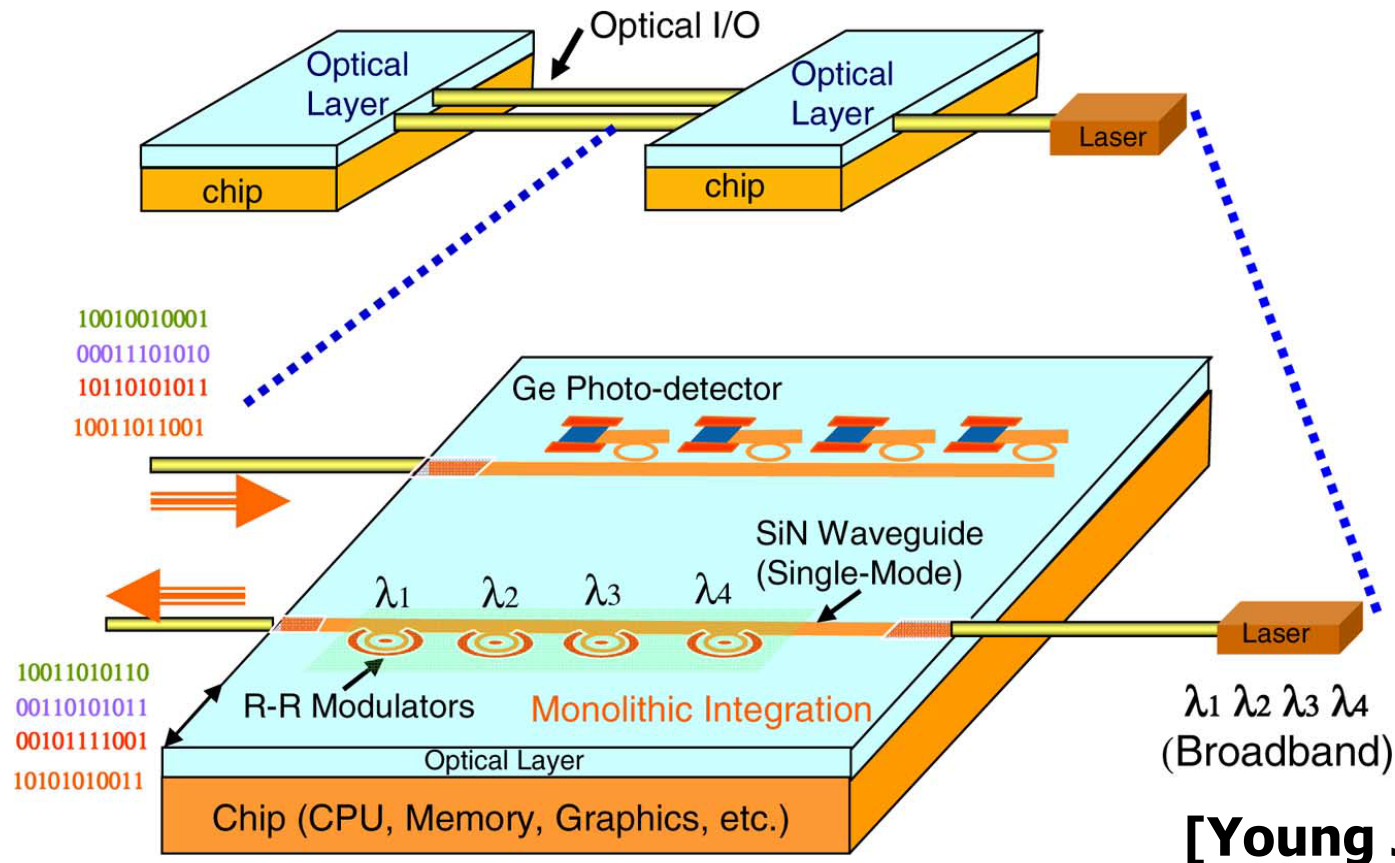


[Lau JEP 2025]



- 4x28.8Tb/s Switch
 - Based on microring resonator modulators driven by 200Gb/s PAM4 SERDES
 - 3D hybrid bonding for EIC-PIC interface
 - External laser source
 - 6 optical assemblies X 3 photonic engines
 - 288 data link fibers per 28.8Tb/s plus 36 laser fibers

Wavelength-Division Multiplexing



- WDM allows for multiple high-bandwidth (10+Gb/s) signals to be packed onto one optical channel

Optical Interconnects

- Electrical Channel Issues
- Optical Channel
- Optical Transmitter Technology
- Optical Receiver Technology
- Optical Integration Approaches

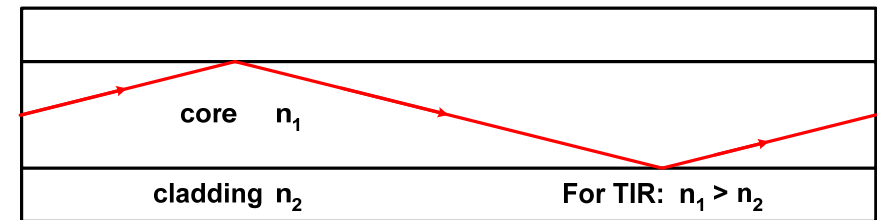
Optical Channels

- Short distance optical I/O channels are typically either waveguide (fiber)-based or free-space
- Optical channel advantages
 - Much lower loss
 - Lower cross-talk
 - Smaller waveguides relative to electrical traces
 - Potential for multiple data channels on single fiber via WDM

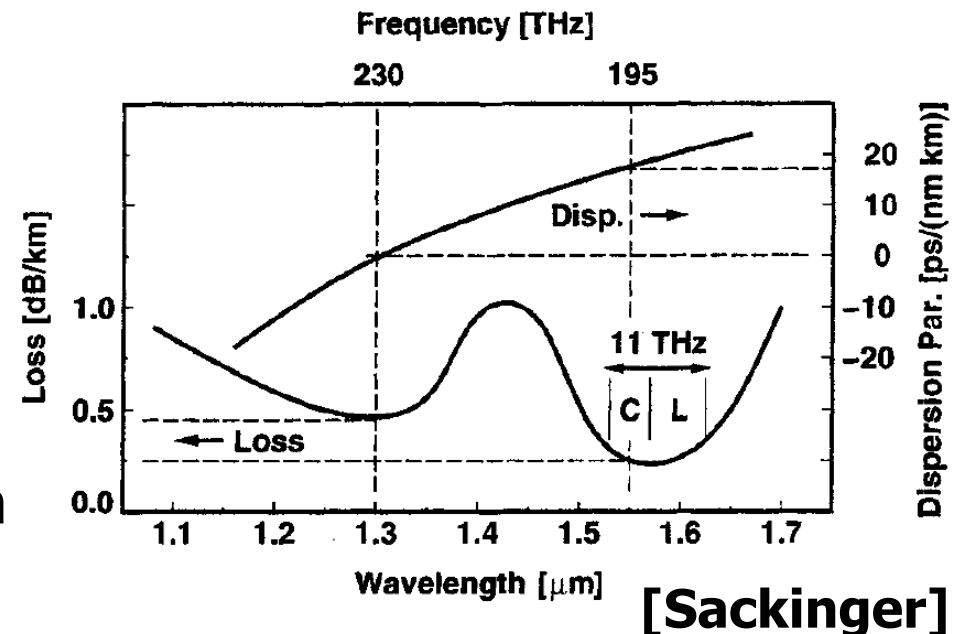
Waveguide (Fiber)-Based Optical Links

- Optical fiber loss is specified in dB/km
 - Single-Mode Fiber loss $\sim 0.25\text{dB/km}$ at 1550nm
 - RF coaxial cable loss $\sim 100\text{dB/km}$ at 10GHz
- Frequency dependent loss is very small
 - $< 0.5\text{dB/km}$ over a bandwidth $> 10\text{THz}$
- Bandwidth may be limited by dispersion (pulse-spreading)
 - Important to limit laser linewidth for long distances ($> 1\text{km}$)

Optical Fiber Cross-Section

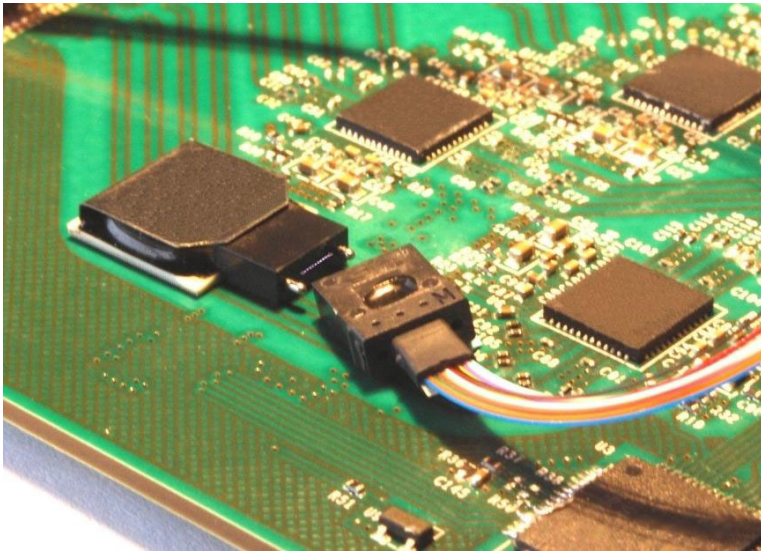


Single-Mode Fiber Loss & Dispersion



Inter-Chip Waveguide Examples

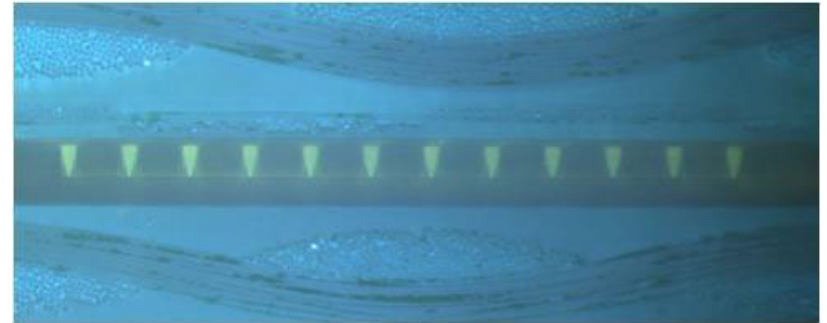
12-Channel Ribbon Fiber



[Reflex Photonics]

12 channels at a $250\mu\text{m}$ pitch
10Gb/s mod. $\rightarrow$ 40Gb/s/mm

Optical Polymer Waveguide in PCB



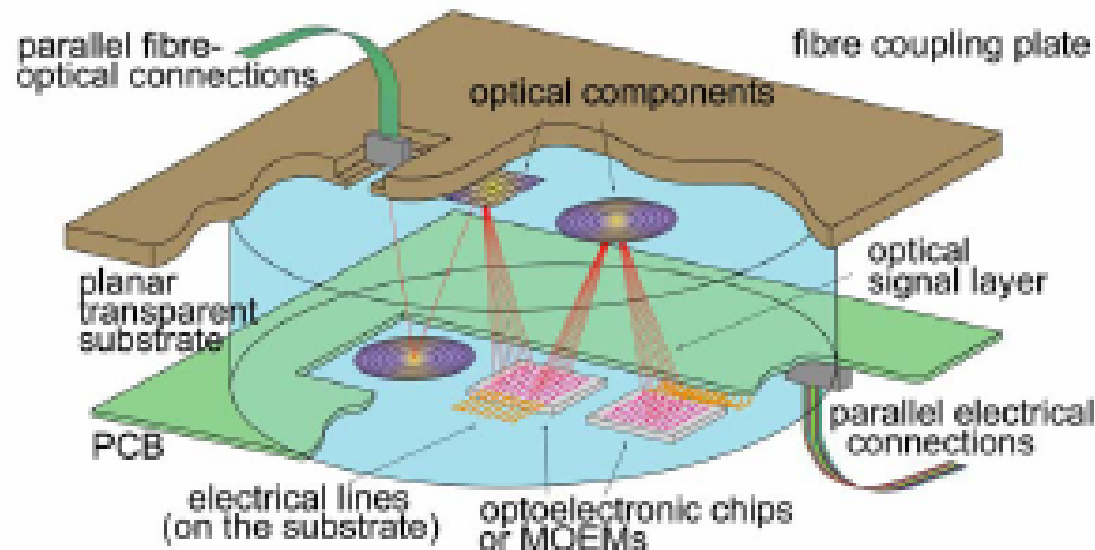
[Immonen 2009]

$<100\mu\text{m}$ channel pitch possible
10Gb/s mod. $\rightarrow$ 100+Gb/s/mm

- Typical differential electrical strip lines are at $\sim 500\mu\text{m}$ pitch

Free-Space Optical Links

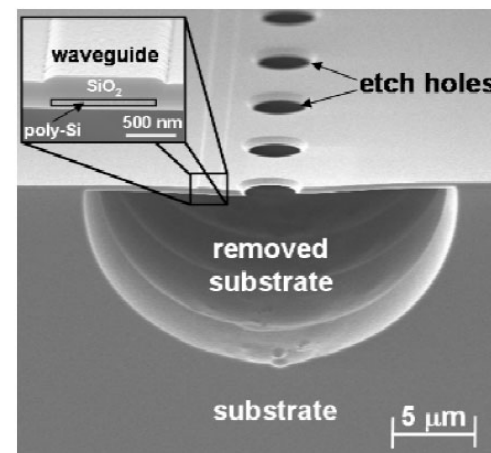
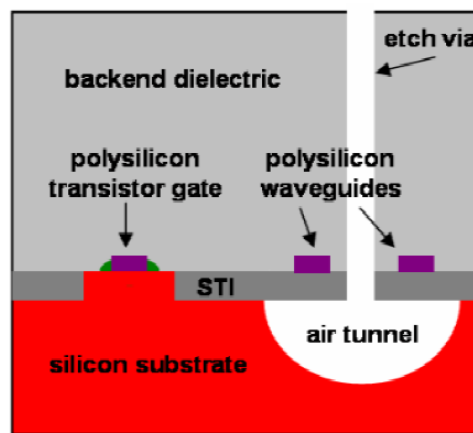
[Gruber]



- Free-space (air or glass) interconnect systems have also been proposed
- Optical imaging system routes light chip-to-chip

CMOS Waveguides – Bulk CMOS

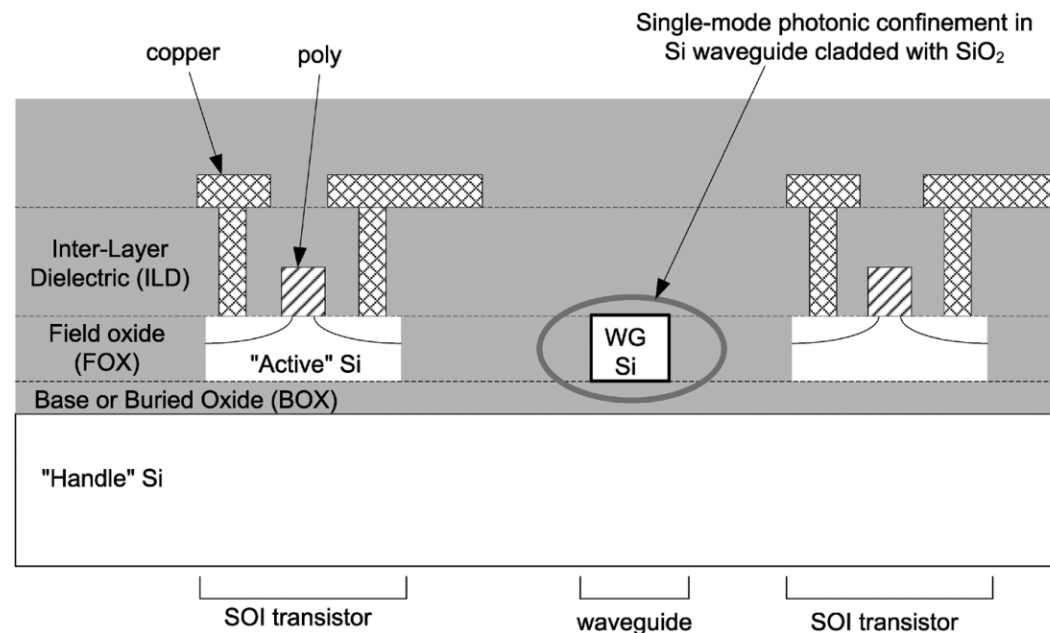
- Waveguides can be made in a bulk process with a polysilicon core surrounded by an SiO₂ cladding
- However, thin STI layer means a significant portion of the optical mode will leak into the Si substrate, causing significant loss (1000dB/cm)
- Significant post-processing is required for reasonable loss (10dB/cm) waveguides in a bulk process



[Holzwarth CLEO 2008]

CMOS Waveguides – SOI

- SOI processes have thicker buried oxide layers to sufficiently confine the optical mode
- Allows for low-loss waveguides

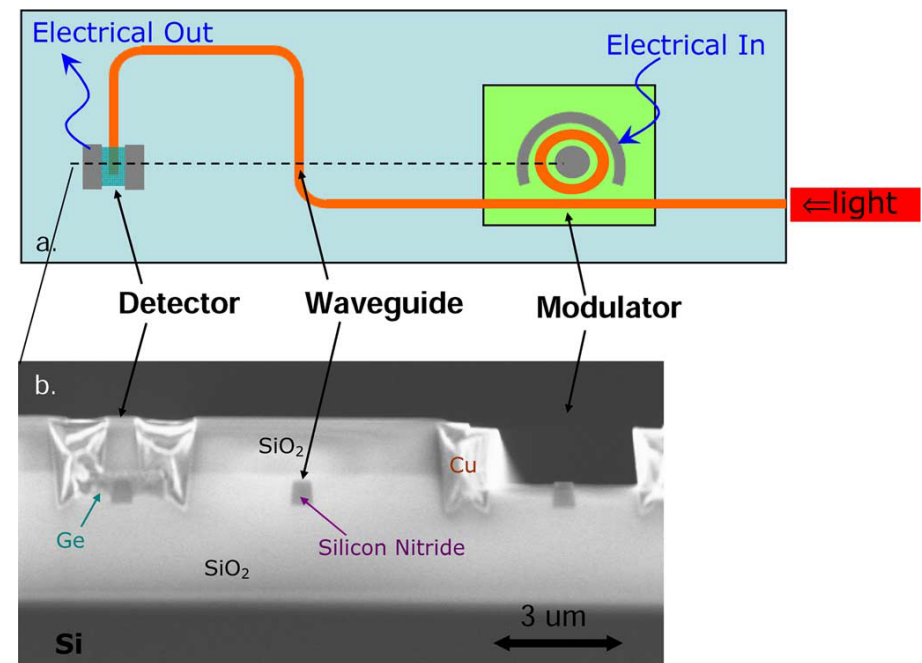


[Narasimha JSSC 2007]

CMOS Waveguides – Back-End Processing

- Waveguides & optical devices can be fabricated above metallization
- Reduces active area consumption
- Allows for independent optimization of transistor and optical device processes

[Young JSSC 2010]

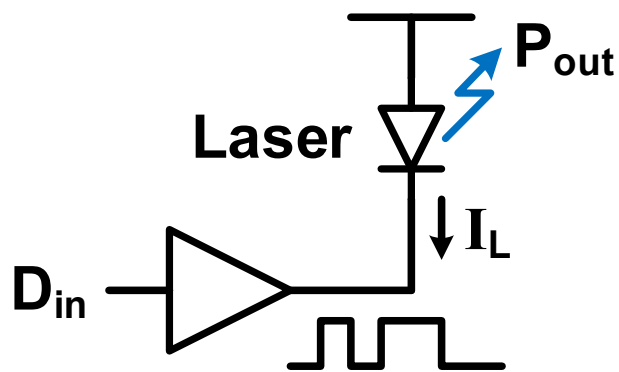


Optical Interconnects

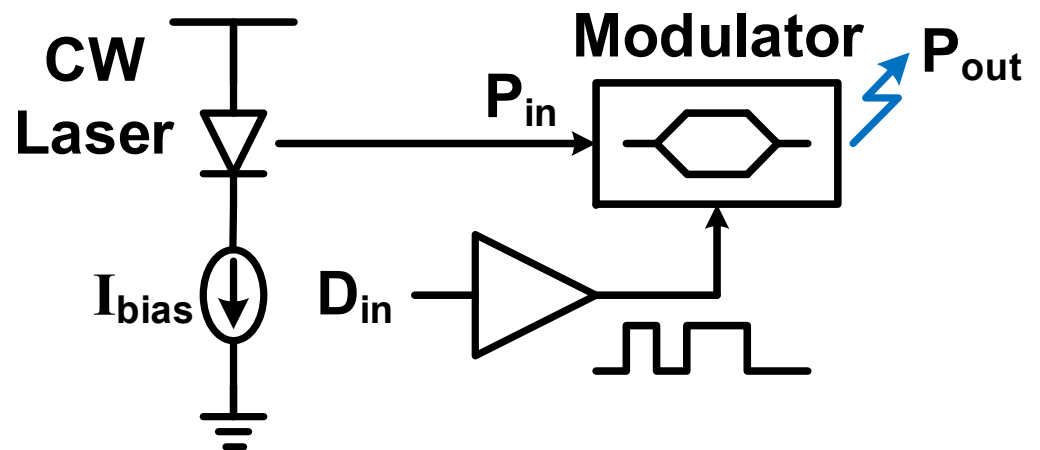
- Electrical Channel Issues
- Optical Channel
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Optical Modulation Techniques

Direct Modulation

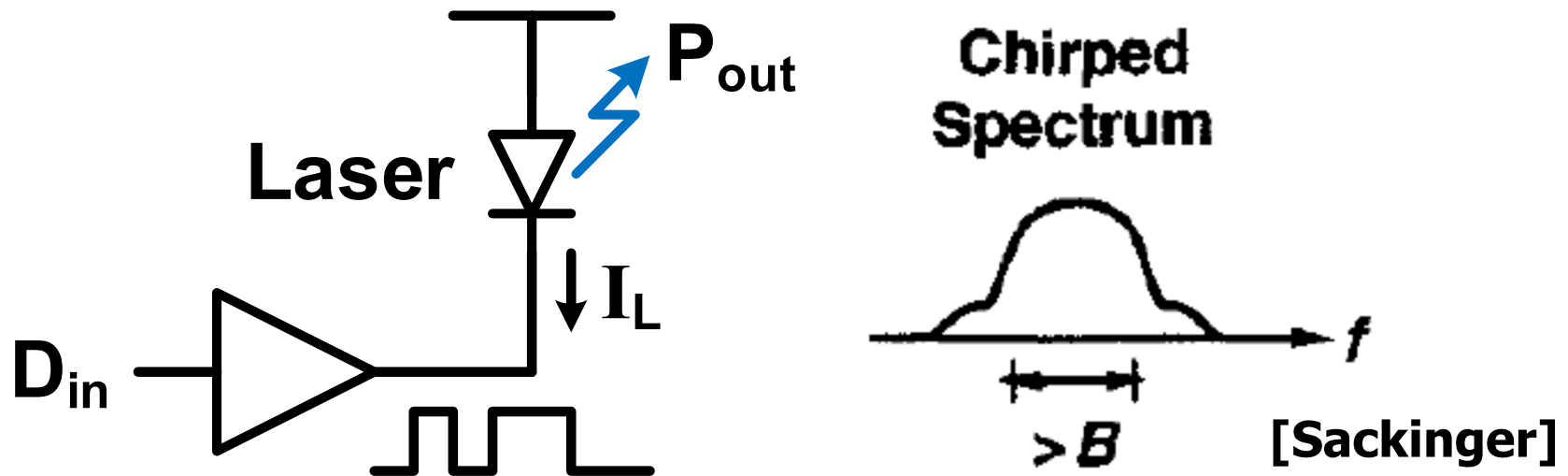


External Modulation



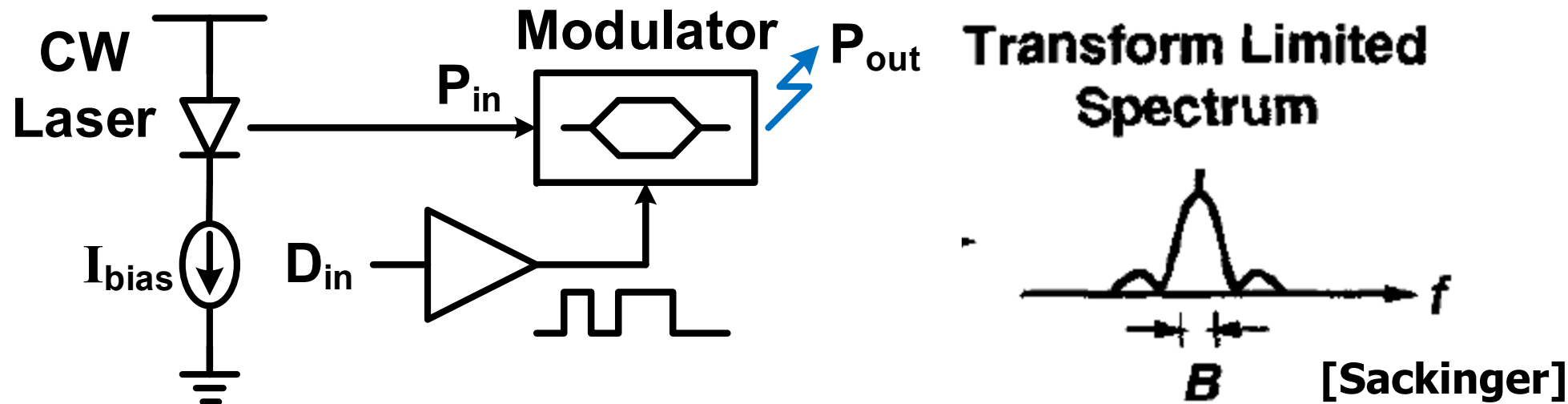
- Two modulation techniques
 - Direct modulation of laser
 - External modulation of continuous-wave (CW) “DC” laser with absorptive or refractive modulators

Directly Modulated Laser



- Directly modulating laser output power
- Simplest approach
- Introduces laser "chirp", which is unwanted frequency (wavelength) modulation
- This chirp causes unwanted pulse dispersion when passed through a long fiber

Externally Modulated Laser



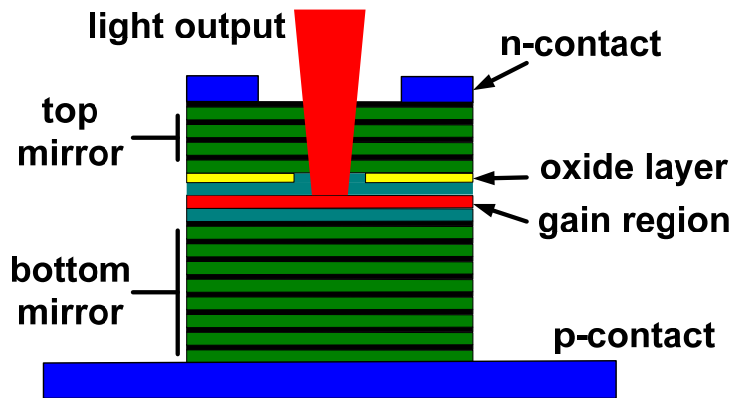
- External modulation of continuous-wave (CW)
“DC” laser with absorptive or refractive modulators
 - Adds an extra component
 - Doesn't add chirp, and allows for a transform limited spectrum

Optical Sources for Chip-to-Chip Links

- Vertical-Cavity Surface-Emitting Laser (VCSEL)
- Electro-Absorption Modulator (EAM)
- Ring-Resonator Modulator (RRM)
- Mach-Zehnder Modulator (MZM)

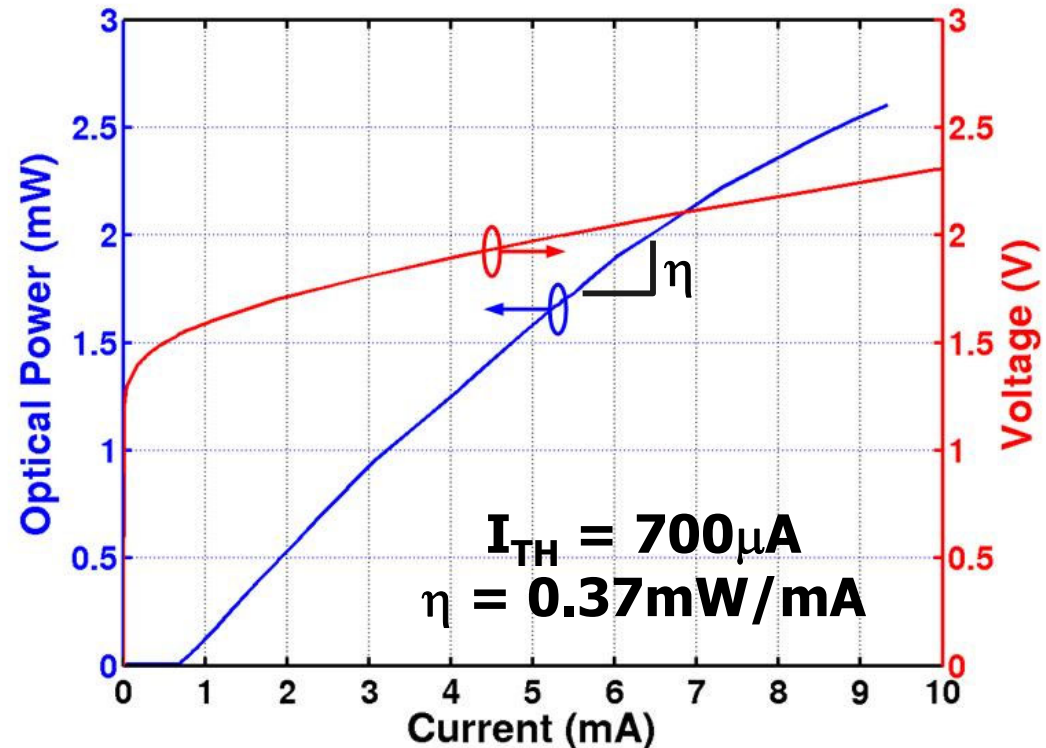
Vertical-Cavity Surface-Emitting Laser (VCSEL)

VCSEL Cross-Section



- VCSEL emits light perpendicular from top (or bottom) surface
- Important to always operate VCSEL above threshold current, I_{TH} , to prevent “turn-on delay” which results in ISI
- Operate at finite extinction ratio (P_1/P_0)

VCSEL L-I-V Curves



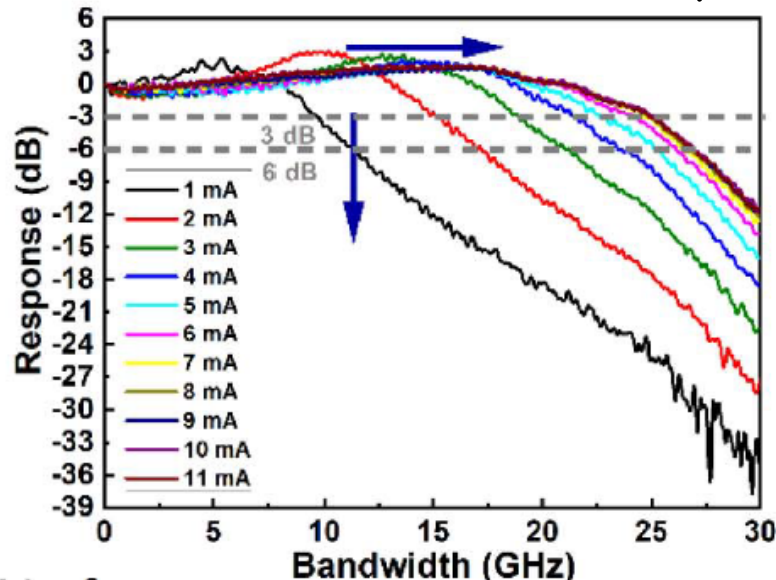
$$P_o = \eta(I - I_{TH})$$

$$\text{Slope Efficiency } \eta = \frac{\Delta P}{\Delta I} \left(\frac{\text{W}}{\text{A}} \right)$$

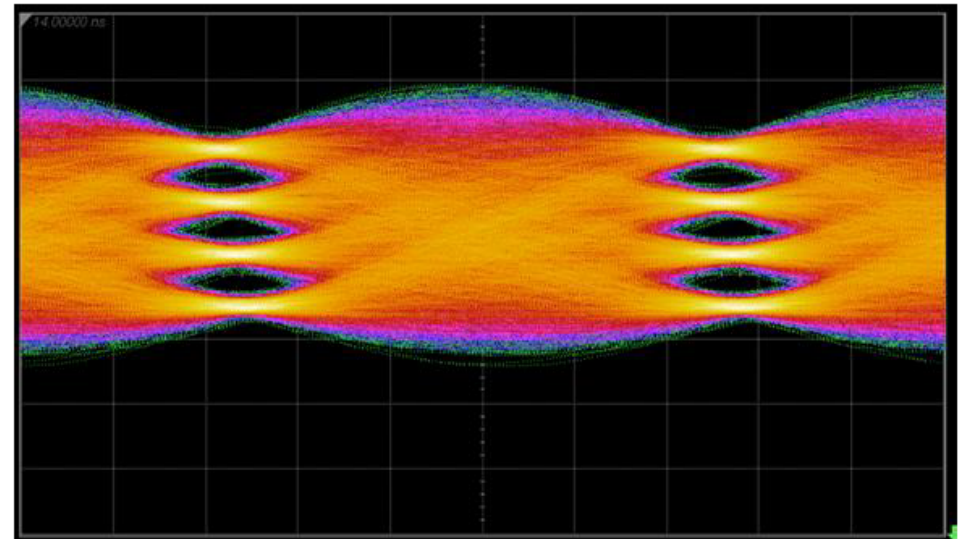
VCSEL Bias-Dependent Bandwidth

[Wu PhRes 2024]

$$BW \propto \sqrt{I_{avg} - I_{TH}}$$



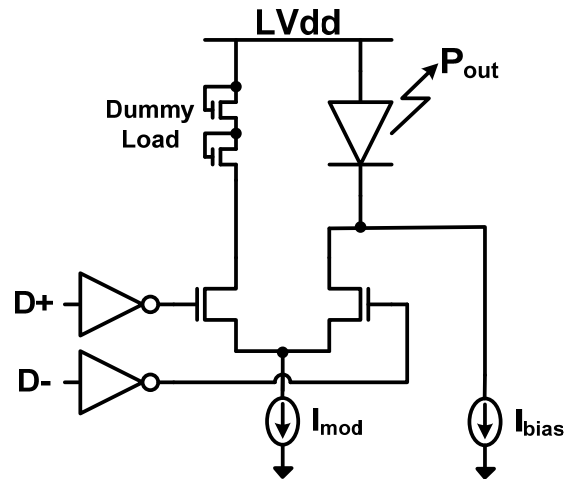
200Gb/s 850nm MM-VCSEL
(c) 9mA, 50m (EMB 7500 MHz.km)



- VCSEL electrical impedance and rate equations cause bias-dependent bandwidth
 - State of the art is 39GHz at 9mA bias [Wang PhWest 2025]
- Lower bias results in an underdamped low-bandwidth response
- Driving the VCSELs significantly above threshold reduces ER and results in a speed/power tradeoff
- 200Gb/s PAM4 demonstrated

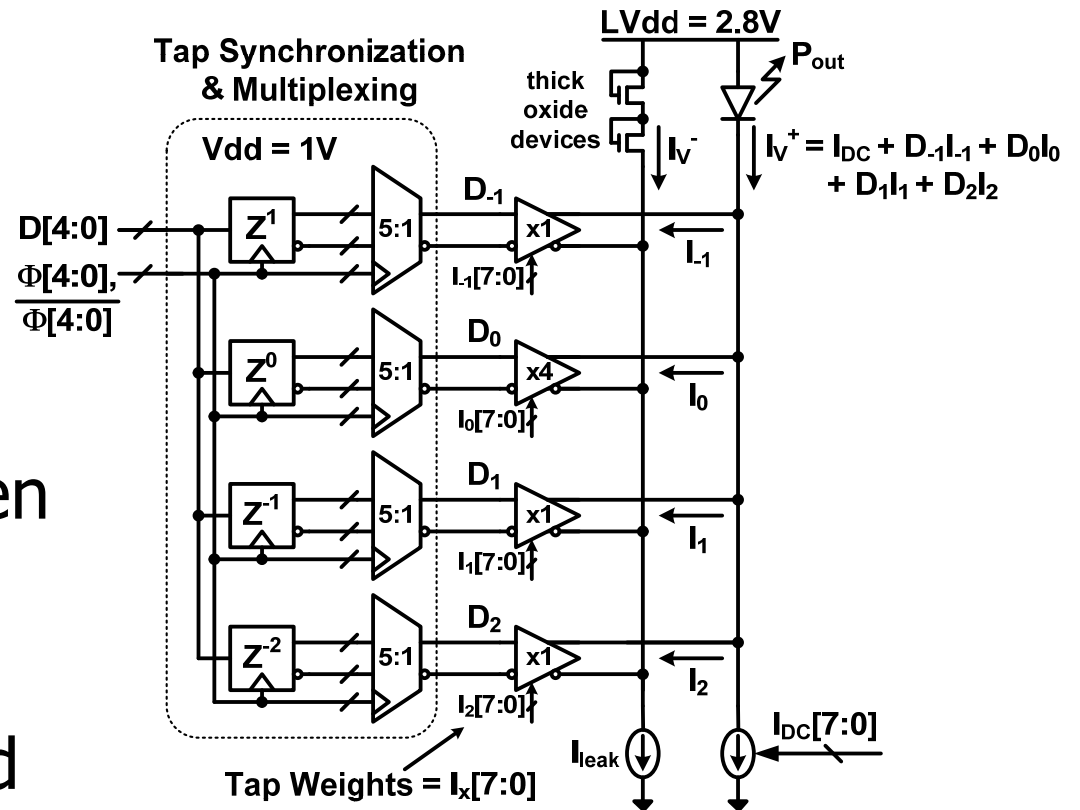
VCSEL Drivers

Current-Mode VCSEL Driver



- Current-mode drivers often used due to linear L-I relationship
- Equalization can be added to extend VCSEL bandwidth for a given current density

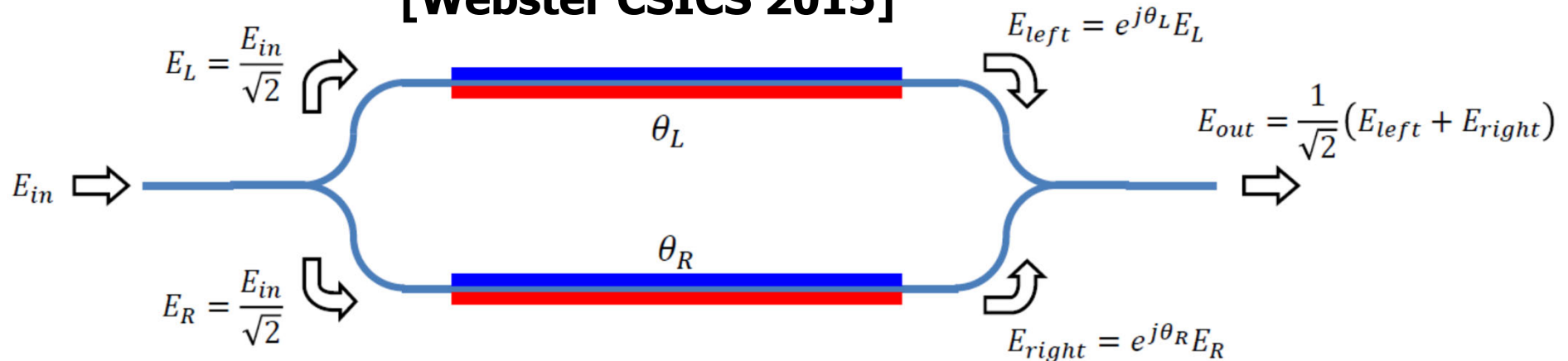
VCSEL Driver w/ 4-tap FIR Equalization



S. Palermo and M. Horowitz, "High-Speed Transmitters in 90nm CMOS for High-Density Optical Interconnects," *ESSCIRC*, 2006.

Mach-Zehnder Modulator (MZM)

[Webster CSICS 2015]



- An optical interferometer is formed with the incoming light split, experiencing phase shifts through the two paths, and then recombined
- Assuming no loss and a perfect 50/50 splitter/combiner

$$\Delta\phi = \frac{(\theta_R - \theta_L)}{2} \quad \phi = \frac{(\theta_R + \theta_L)}{2}$$

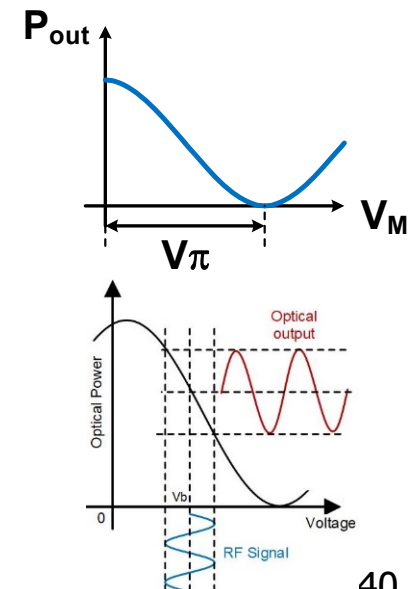
Field Response

$$E_{out} = E_{in} \cos(\Delta\phi) e^{j\phi}$$

Intensity Response

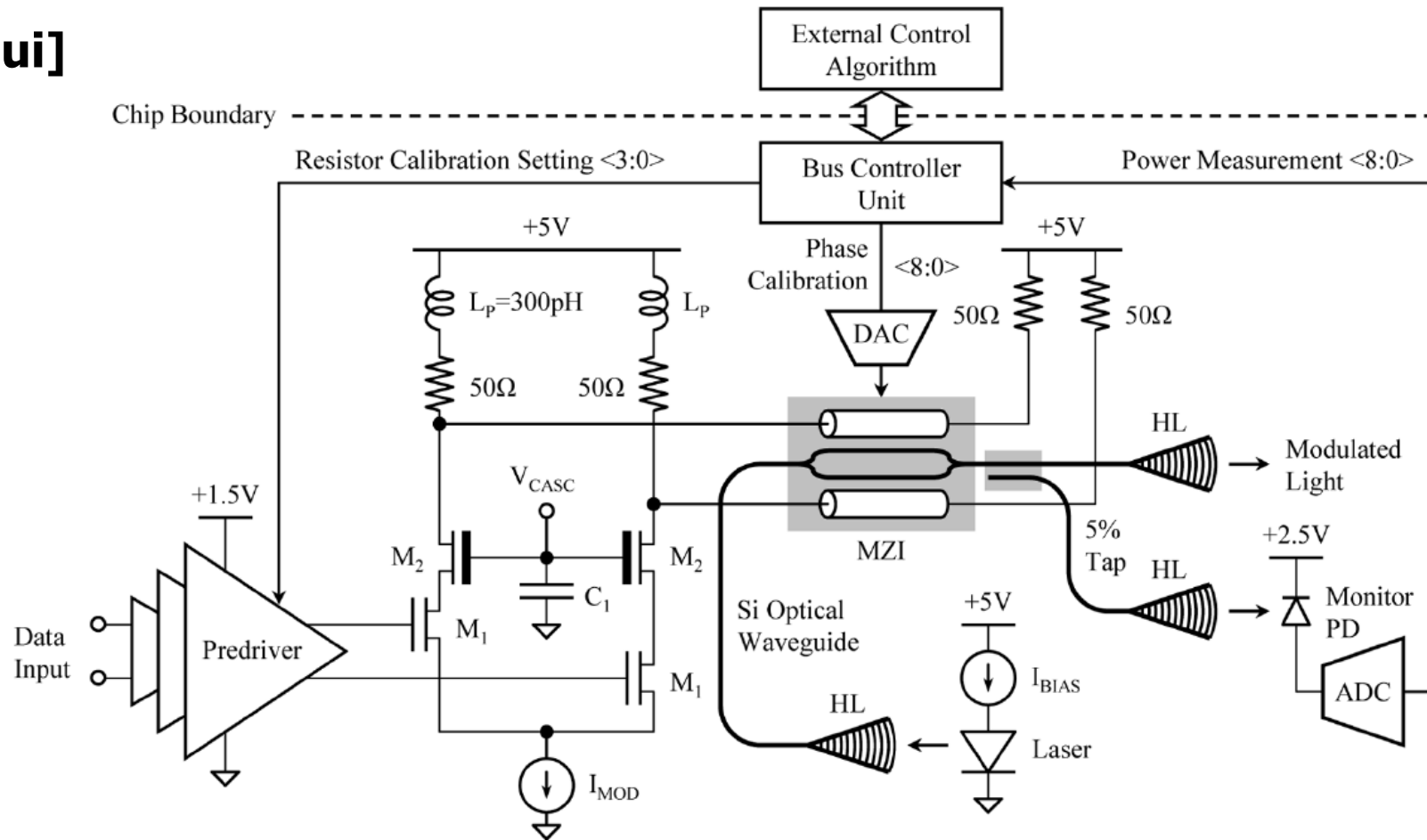
$$P_{out} = |E_{out}|^2 = \frac{1}{2} |E_{in}|^2 [1 + \cos(\theta_R - \theta_L)]$$

$$\frac{P_{out}}{P_{in}} = \frac{1}{2} [1 + \cos(\theta_R - \theta_L)] = \frac{1}{2} \left[1 + \cos\left(\pi \cdot \frac{V_M}{V\pi}\right) \right]$$



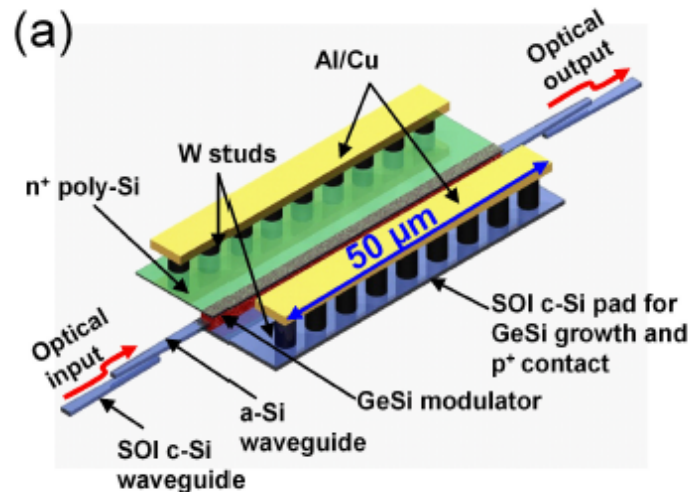
MZM Driver

[Analui]

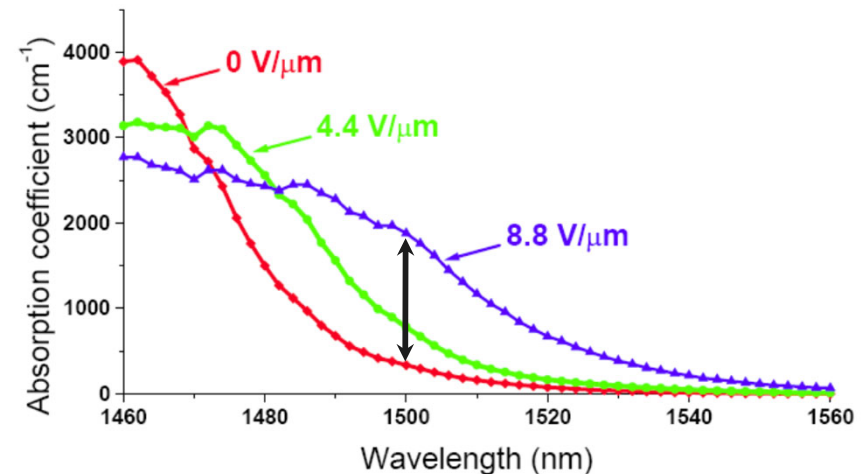


- Long device (several mm) requires driver to drive low-impedance transmission line at potentially high swing ($5V_{ppd}$)
- While much higher power relative to RRM, they are less sensitive to temperature variations

Electro-Absorption Modulator (EAM)



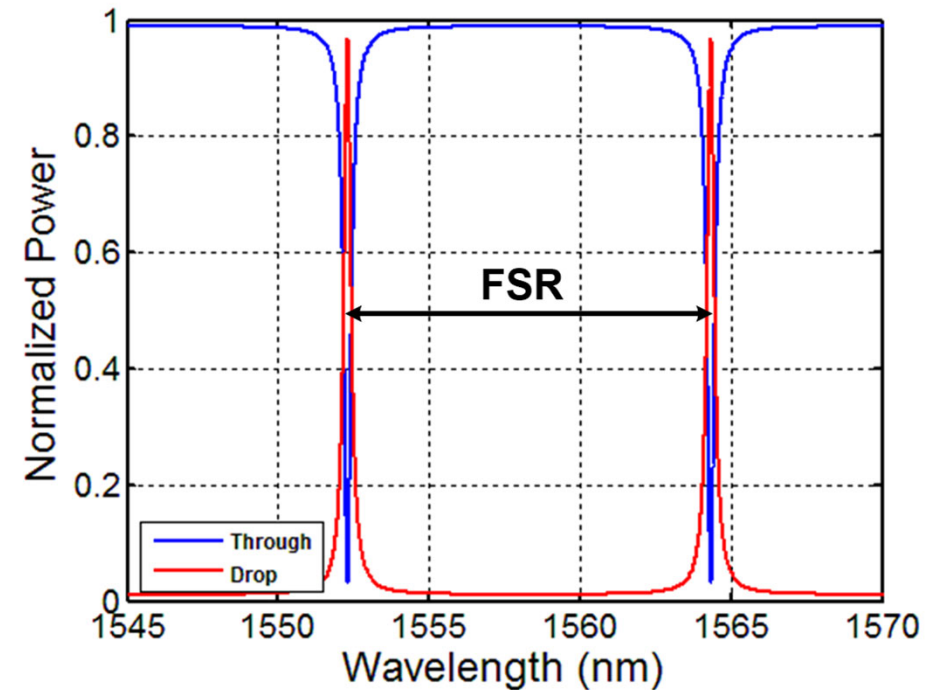
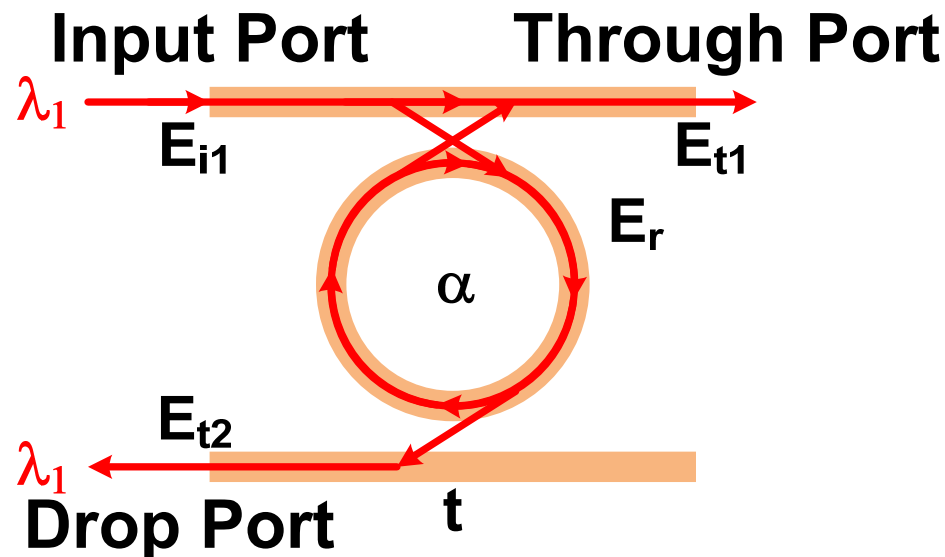
Waveguide EAM [Liu 2008]



[Helman JSTQE 2005]

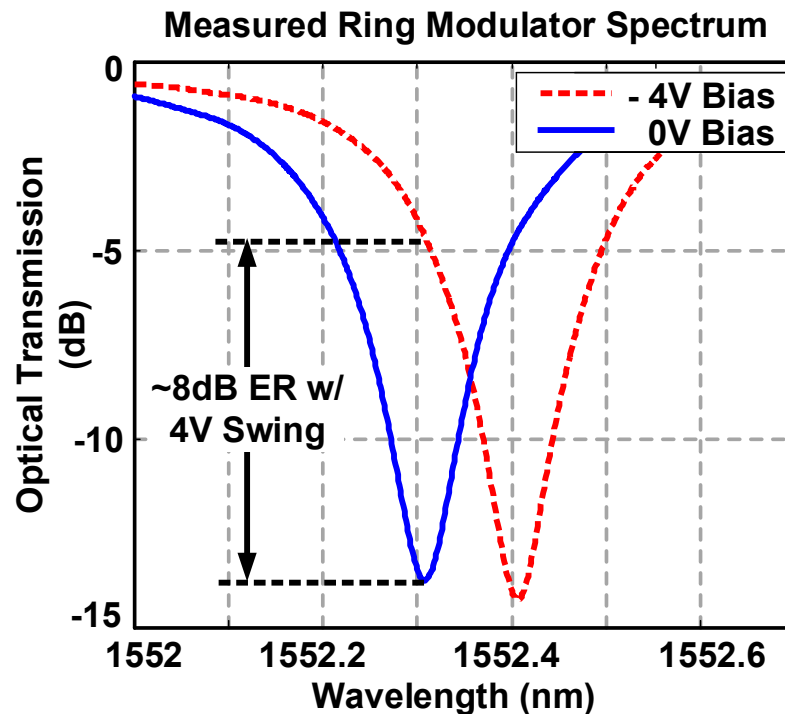
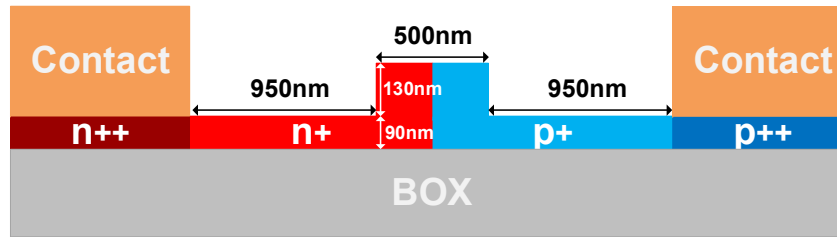
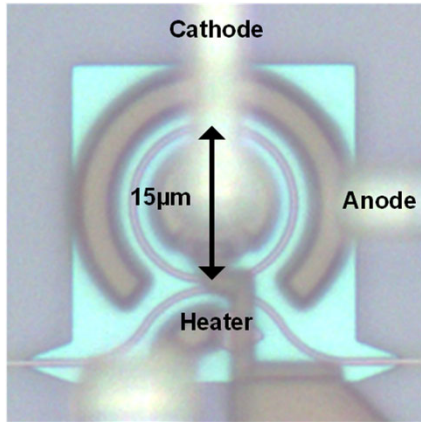
- Electro-absorption modulators operate with voltage-dependent absorption of light passing through the device
- The device structure is a reverse-biased p-i-n diode
- The Franz-Keldysh effect describes how the effective bandgap of the semiconductor decreases with increasing electric field, shifting the absorption edge
- While this effect is weak, it can be enhanced with device structures with multiple quantum wells (MQW) through the quantum-confined Stark effect

Ring Resonator Filter



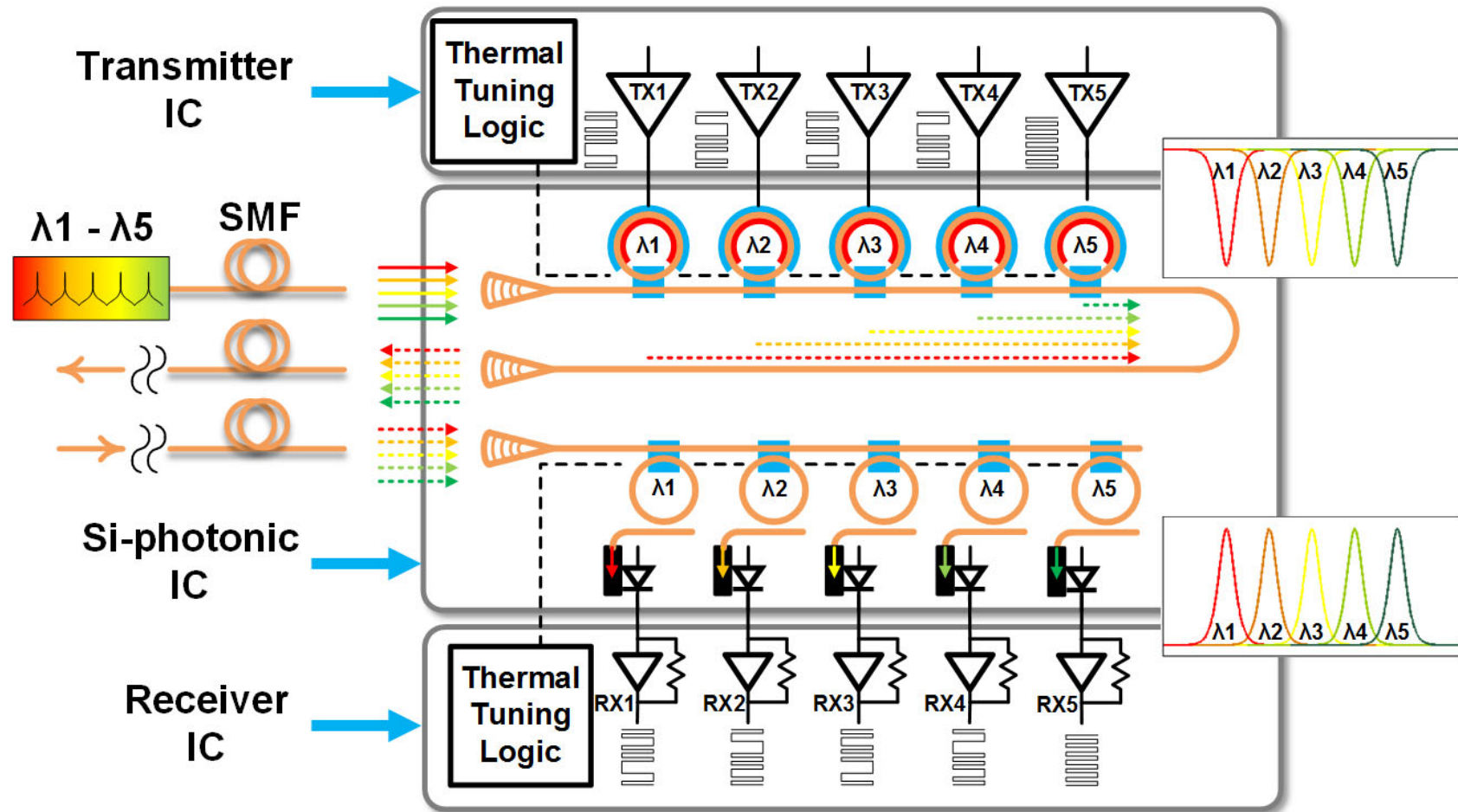
- Ring resonators display a high- Q notch filter response at the through port and a band-pass response at the drop port
- This response repeats over a free spectral range (FSR)

Carrier-Depletion Ring Modulator



	ISSCC 2013	ISSCC 2015
Ring Type	Injection	Depletion
Doping Profile	PIN	Lateral PN
Q	8000	5000
Tunability (pm/V)	350	25
Data Rate	9Gb/s	25Gb/s
Swing for >7dB ER	< 2V _{pp}	> 4V _{pp}

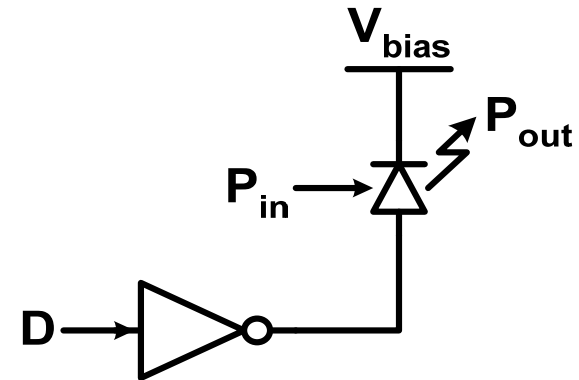
Wavelength Division Multiplexing w/ Ring Resonators



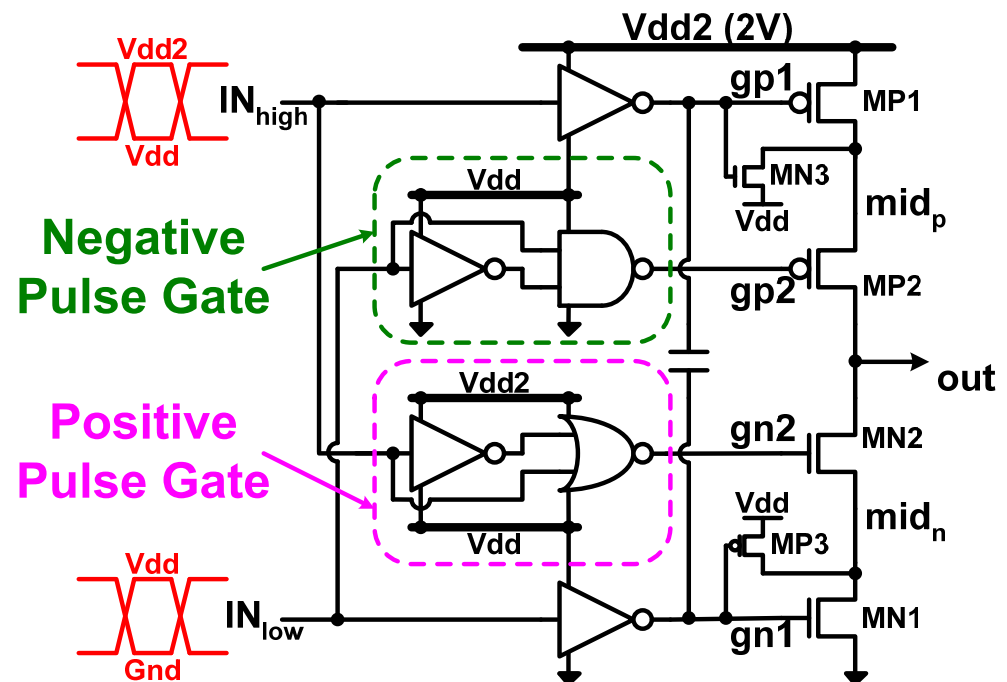
- Ring resonators can act as both modulators and add/drop filters to steer light to receivers or switch light to different waveguides
- Potential to pack >100 waveguides, each modulated at more than 10Gb/s on a single on-chip waveguide

CMOS Modulator Driver

- Simple CMOS-style voltage-mode drivers can drive EAM and RRM due to their small size
- Device may require swing higher than nominal CMOS supply
 - Pulsed-Cascode driver can reliably provide swing of $2xV_{dd}$ (or $4xV_{dd}$) at up to $2F_{O4}$ data rate



Pulsed-Cascode Driver

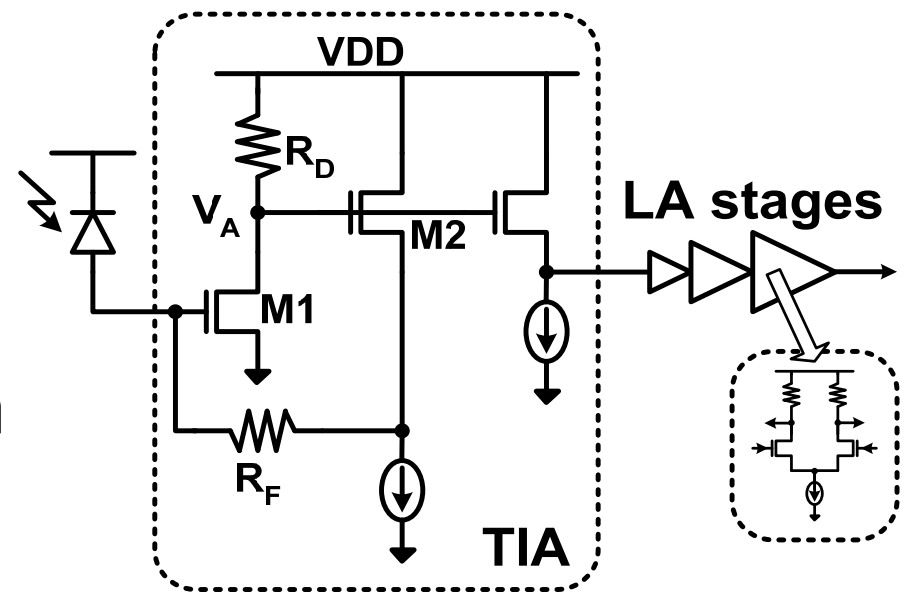


Optical Interconnects

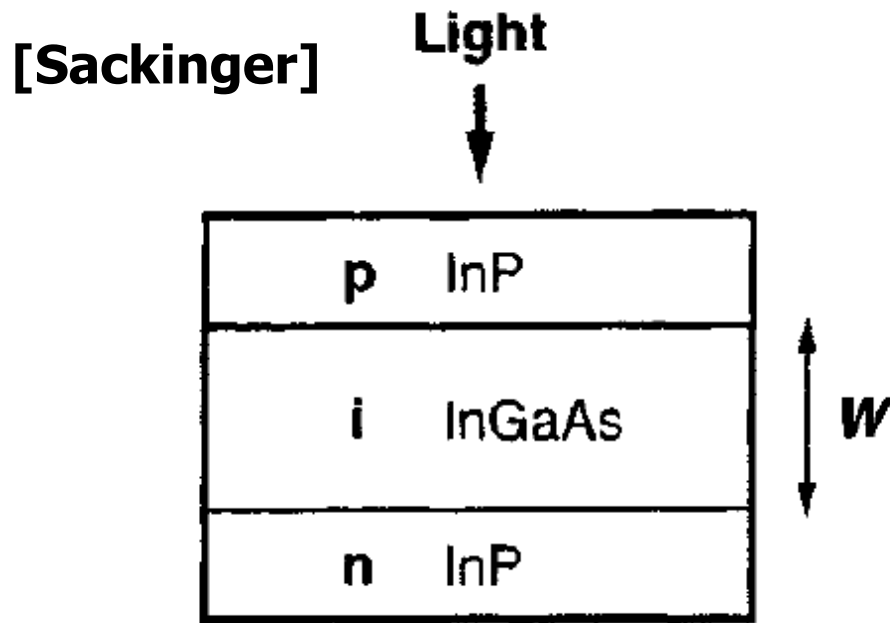
- Electrical Channel Issues
- Optical Channel
- Optical Transmitter Technology
- Optical Receiver Technology
- Optical Integration Approaches

Optical Receiver Technology

- Photodetectors convert optical power into current
 - p-i-n photodiodes
 - Integrated metal-semiconductor-metal photodetector
- Electrical amplifiers then convert the photocurrent into a voltage signal
 - Transimpedance amplifiers
 - Limiting amplifiers
 - Integrating optical receiver



p-i-n Photodiode



Responsivity:

$$\rho = \frac{I}{P_{opt}} = \frac{\eta_{pd} \lambda q}{hc} = 8 \times 10^5 (\eta_{pd} \lambda) \quad (\text{mA/mW})$$

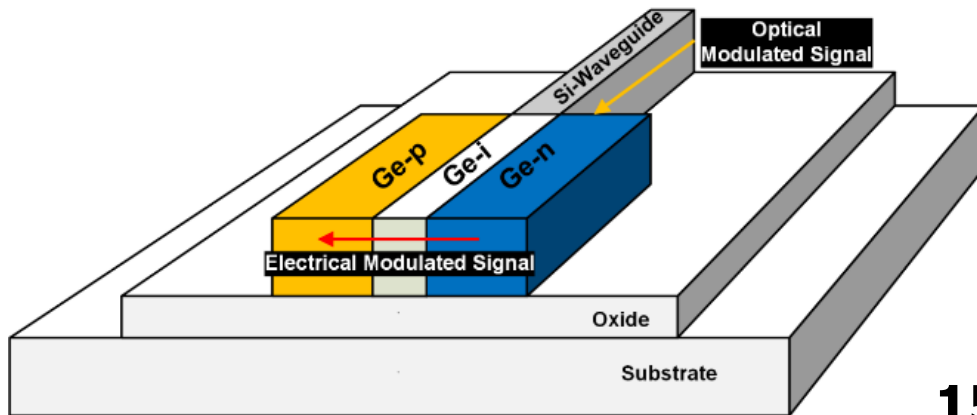
Quantum Efficiency: $\eta_{pd} = 1 - e^{-\alpha W}$

Transit-Time Limited Bandwidth:

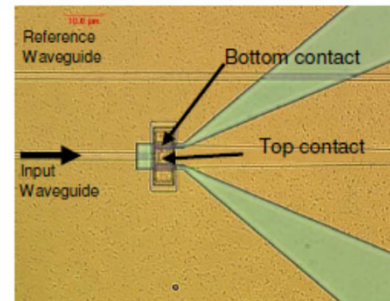
$$f_{3dBPD} = \frac{2.4}{2\pi\tau_{tr}} = \frac{0.45v_{sat}}{W}$$

- Normally incident light absorbed in intrinsic region and generates carriers
- Trade-off between capacitance and transit-time
- Typical capacitance between 100-300fF

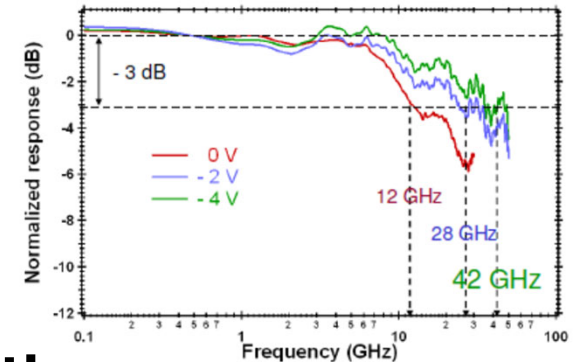
Waveguide p-i-n Photodetector



[Vivien OptExp 2009]



15 μm detector length



- A waveguide p-i-n photodetector structure allows this efficiency-speed trade-off to be broken
- The light travels horizontally down the intrinsic region and the electric field is formed orthogonal
- Allows for both a thin i-region for short transit times and a sufficiently long i-region for high quantum efficiency

Optical Interconnects

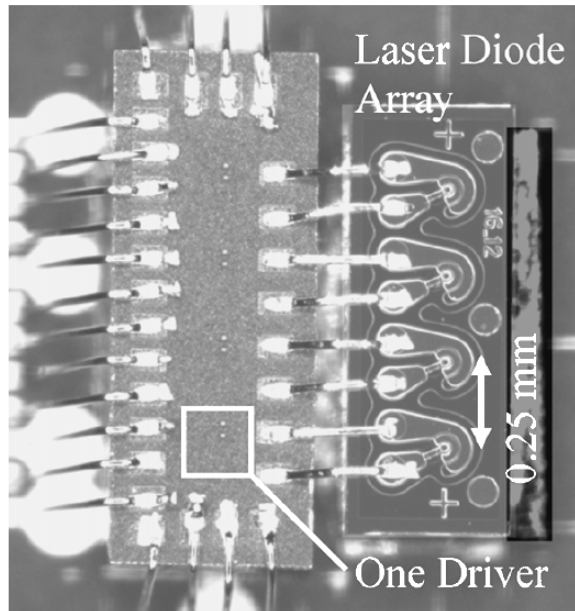
- Electrical Channel Issues
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Optical Integration Approaches

- Efficient cost-effective optical integration approaches are necessary for optical interconnects to realize their potential for improved power efficiency at higher data rates
- Hybrid integration
 - Optical devices fabricated on a separate substrate
- Integrated CMOS photonics
 - Optical devices part of CMOS chip

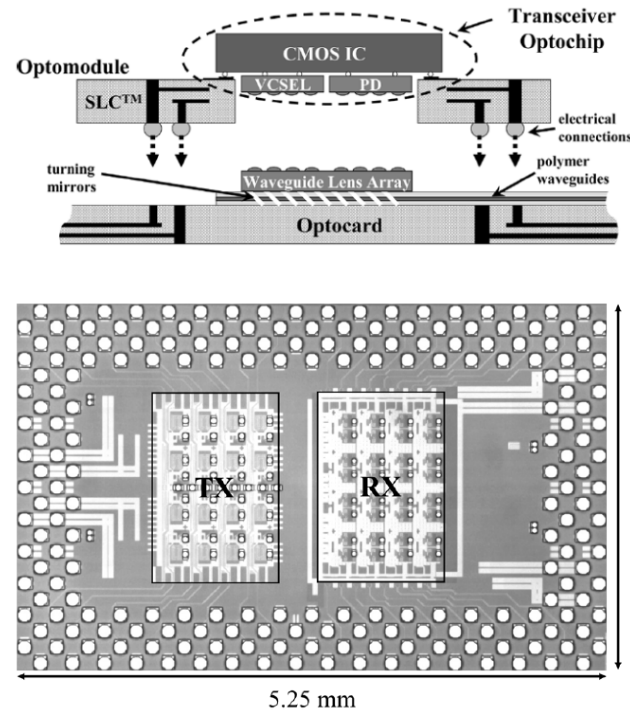
Hybrid Integration

[Kromer]



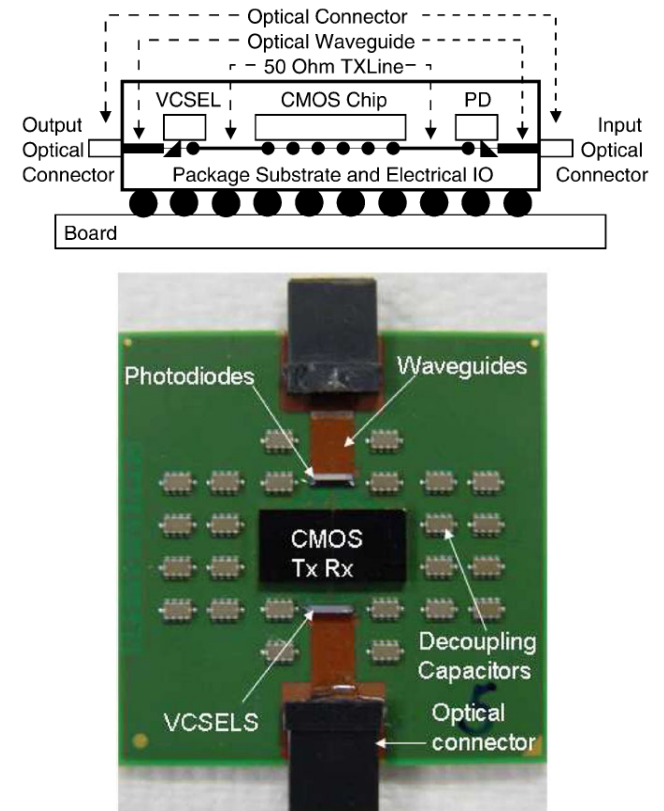
Wirebonding

[Schow]



Flip-Chip Bonding

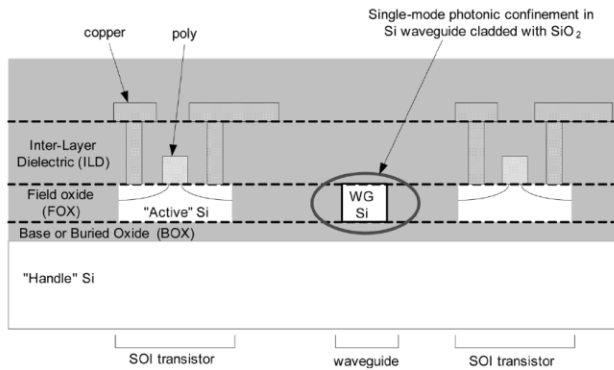
[Mohammed]



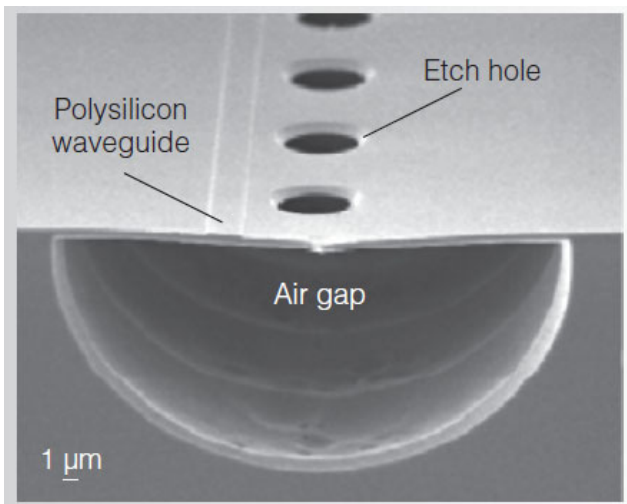
Short In-Package Traces

Integrated CMOS Photonics

SOI CMOS Process [Analui]



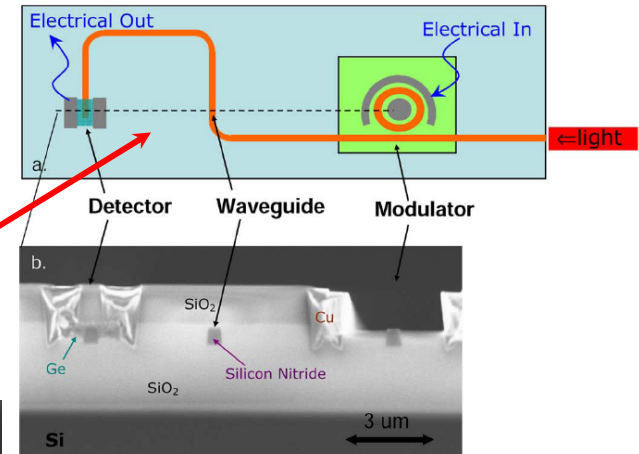
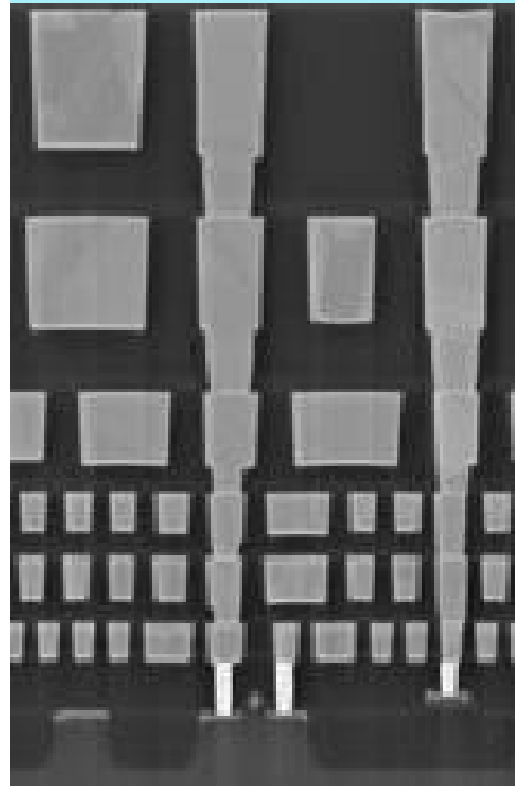
Bulk CMOS Process



[Batten]

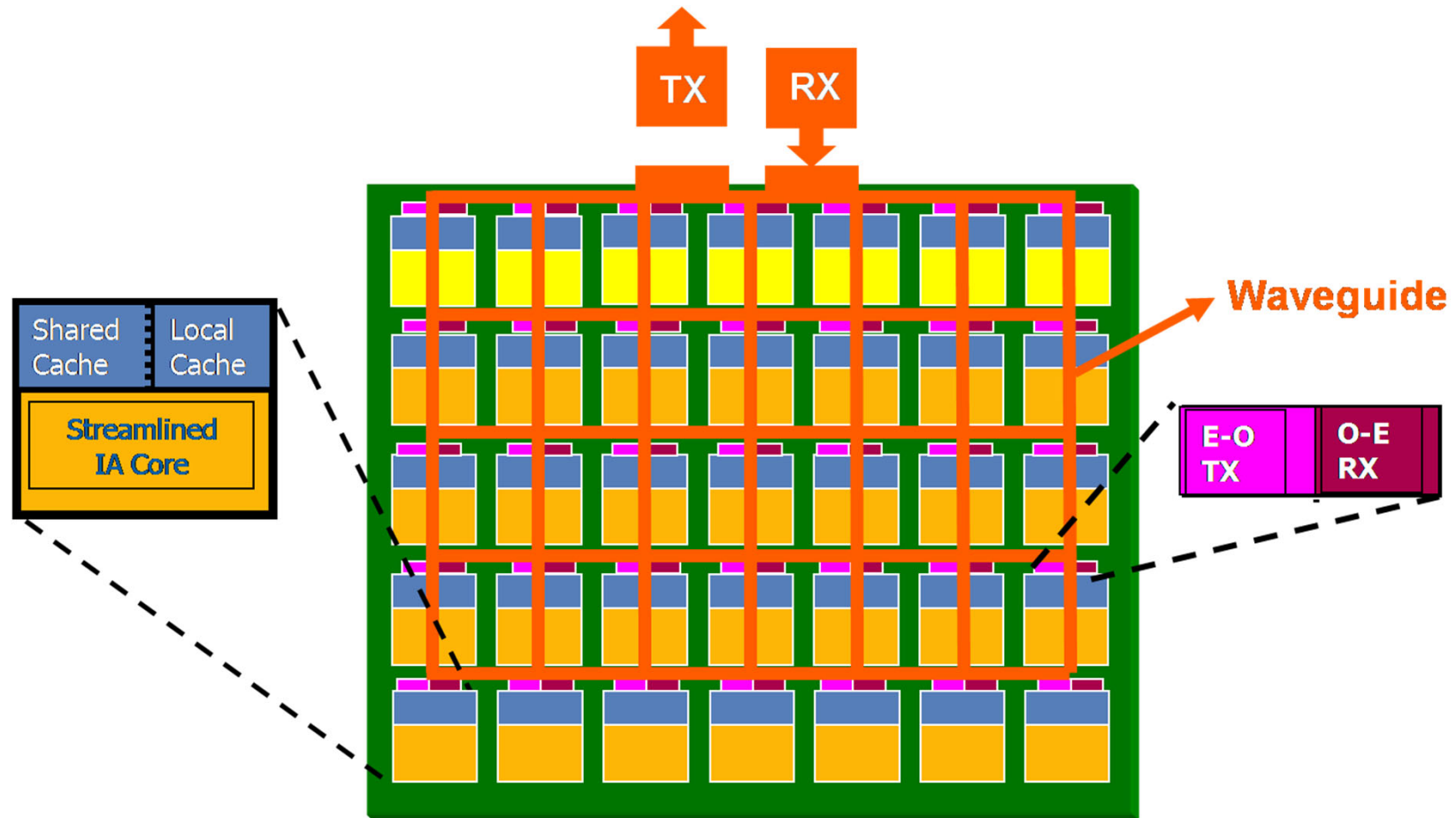
"Optics On Top"

Optical Layer



[Young]

Future Photonic CMOS Chip



- Unified optical interconnect for on-chip core-to-core and off-chip processor-to-processor and processor-to-memory

Next Time

- Optical Channels
 - Sackinger Chapter 2